TS2000
TS2100
Robot Controller

INSTRUCTION MANUAL

ARC CONVEYOR

SYNCHRONIZATION FUNCTION

Rev.3
April 1, 2020

Notice

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Shibaura Machine

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Introduction

This manual describes the outline, specifications and commands of the conveyor synchronization function used for SHIBAURA MACHINE's Industrial Robot Controller TS2000.

Section 1 General Descriptions
The outline of this function is described.

Section 2 Specifications
The basic specifications of this function are described.

Section 3 Parameters
The parameter setting procedures for using this function are described.

Section 4 Commands
The robot language of this function is described.

Section 5 Interface
The preparations and cautions for using this function are described.

Section 6 Programming and Teaching
The teaching procedures for using this function are described.

Section 7 Operation
The operation for stopping and resetting this function is described.

Section 8 Troubleshooting

The contents of this manual are subject to change without prior notice to effect improvements.
Various instruction manuals are available, including the Introductory, Startup, Operation, Robot Language, Interface, Communications, Transportation and Installation, and Maintenance manuals. Refer to these manuals as necessary.

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1. General Descriptions

This section outlines the conveyor synchronization function.
The conveyor synchronization processing unit reads the conveyor present position (speed) from the encoder attached to the conveyor, then starts processing the conveyor synchronization tracking based on the encoder counter at the same time that the trigger sensor has detected a workpiece flowing on the conveyor.

[1] Firstly, the conveyor rotation center position (X, Y) is figured out as seen from the origin of the world coordinate system.

[2] Additionally, points A (synchronization start point) and B (synchronization end point) in Fig. 1.1 are taught on the conveyor to specify the synchronization direction and synchronization travel distance of the conveyor. Thus, the robot moves synchronously along an arc connecting point A and point B in the conveyor distance of "B–A".

[3] <u>If the robot waiting position for conveyor synchronization is required as shown in the figure below, [P0: Robot waiting position] should be provided.</u>

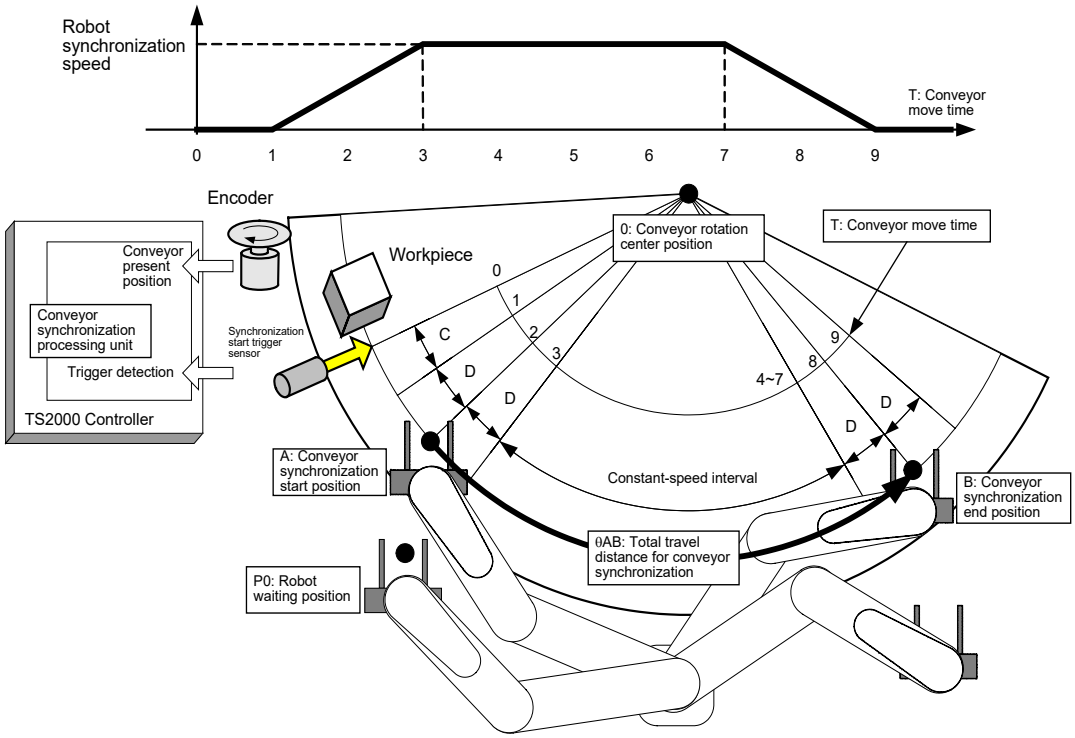


Fig. 1.1 Outline of synchronized operation (*)

[1] Conveyor move time T (0 ~ 1): Delay interval
The robot stops at the waiting position during the time from the trigger signal detection by the controller to the conveyor movement by [C: Conveyor synchronization delay distance].

[2] Conveyor move time T (1 ~ 3): Acceleration interval
If the robot momentarily starts (or stops) movement at the conveyor speed, an excessive impact may be imposed on the robot or workpiece, or the robot will vibrate. To avoid this, [D: Acceleration/deceleration distance for conveyor synchronization] should be specified to assure smooth start and stop of the robot. When T = 1, the robot starts accelerating from point A, and catches up with the conveyor speed when T = 3. In other words, while the conveyor moves by "2×D", the robot accelerates to the conveyor speed in the angle of "D", starting from point A.

[3] Conveyor move time T (3 ~ 7): Constant-speed interval
During conveyor synchronization (i.e., while the robot is moving by tracking the conveyor), the robot moves in parallel with the conveyor at the same speed as the conveyor. During this time, the workpiece is chucked. (Teaching of the synchronized operation program and work positions is performed in this interval.)

[4] Conveyor move time T (7 ~ 9): Deceleration interval
When T = 7, the robot starts decelerating. When T = 9, the robot stops completely at point B. That is, while the conveyor moves by "2D", the robot slows down and stops in the angle of "D".

(*) Both C and D are the independent variables. In the above explanation, C = D to simplify the operation. It is also possible to specify another different values for C and D.

! CAUTION

1. Even if the program stopped during conveyor synchronization, continuous operation is possible by using the restart function just after the program stop. In principle, however, after the stop operation, be sure to reset the operation, move the robot to the safe place by manual guidance, reset the program, then resume the operation. (For details, see Para. 7.2.)
2. Unless an error is caused, DO NOT effect the servo OFF or emergency stop during conveyor synchronization. Otherwise, the robot hand will collide with a workpiece on the synchronized conveyor.
3. To avoid the above accident, construct the system, taking notice of the following matters.
   - [1] If the conveyor or other external equipment enters an emergency stop state, the robot should also enter an emergency stop state.
   - [2] If the robot enters an emergency stop state, the conveyor or other external equipment should also enter an emergency stop state.
   - [3] If the robot program stops due to some reason (error stop/stop operation by operator), the conveyor should be stopped also.
4. Before teaching operation, be sure to stop the conveyor. If the conveyor is moving, the synchronization start trigger sensor may operate under some teaching condition, resulting in an unexpected movement or collision. (The robot at the robot waiting position is only kept stationary until the trigger sensor turns on, and it can start moving at any time.)
5. The robot retract position should be the completely safe position where it will not collide with a workpiece running on the conveyor even if the robot operation is stopped. (Desirably, the robot waiting position should serve as the robot retract position.)
6. The trigger sensor should be adjusted so that it can always be turned on at a predetermined position without causing shift of the workpiece detecting position. If the sensor is turned on due to mixture of different workpieces or at a position other than the adjusted position, the robot tracking shift or collision may be caused.
7. While the test operation mode is selected, DO NOT execute the step of the program (between SYNC command to UNSYNC command), machine lock (dry mode operation) or [MODE: Motion]. If the conveyor is running, an expected collision may be caused.

2. Specifications

This section describes the specifications of the conveyor synchronization function.

Table 2.1 Basic specifications of conveyor synchronization

Item	Specification	Remarks
1. No. of axes (*1)	A maximum of two (2) conveyor axes can be used.	This function can be used at a conveyor speed of 75 ~ 250 mm/sec. The conveyor speed of under 75 mm/sec can be used under some conditions.
2. Synchronous accuracy (*2)	Accuracy: ±10 mm (at 200 mm/sec conveyor speed) • Position interpolation sampling speed: 3 ms • Minimum interpolation unit (X, Y, Z): 0.001 mm • Trigger signal sampling period: 500 μs	The accuracy is the value when the conveyor speed fluctuation is within ±7.5 %.
3. Teaching method	Teaching during conveyor stop.	
4. Robot language	CONV, SYNC, UNSYNC, XIN, CONVSTAT	For details, see Section 4.
5. Conveyor encoder interface (*3)	Power supply: +5 V (supplied from internal source) Method: A, B 90° phase difference Specifications: RS422 differential input (line driver method) Counter: 15-bit absolute counter (0 ~ 32767) Port: Two (2) numbers	Resolution: 10 ~ 100 pulses per 3 msec
6. Trigger signal interface	Power supply: +24 V (supplied from external source) Specification: Photocoupler input Port: Two (2) numbers	

*1: When the conveyor speed is 75 mm/sec or less, an encoder with high resolution is required.

*2: The speed interpolation sampling period can be set by parameter.

*3: The encoder resolution and conveyor stop judgment time can be set by parameters.

3. Parameters

The parameters for the conveyor synchronization function are set in "[U20] Conveyor Parameter" of user parameter file [USER.PAR]. The parameter format is as shown below.

USER.PAR

[U20] Conveyor Parameter

{Conveyor Parameter1(CH1,CH2)}

```
= 0     0     [1] Encoder rotating direction
= 3     3     [2] Encoder sampling period (ms)
= 0     0     [3] Conveyor stop judgment time (ms)
= 0     0     [4] Type of conveyor synchronization
= 0     0     [5] Dummy data
= 0     0     [6] Dummy data
= 0     0     [7] Dummy data
```

{Conveyor Parameter2(CH1,CH2)}

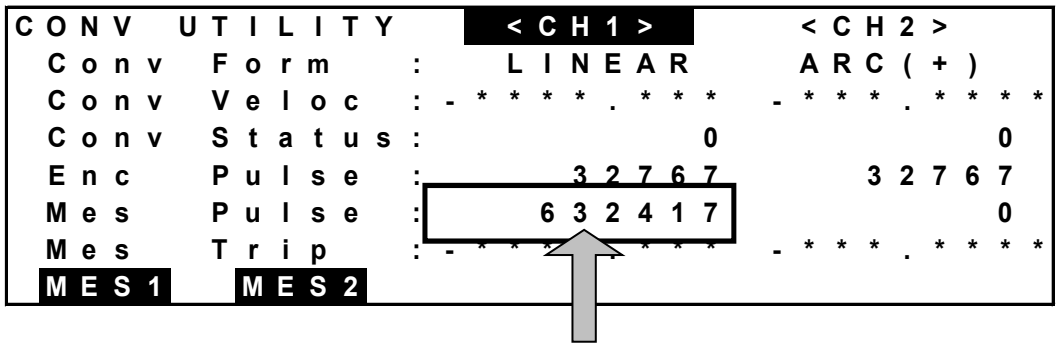
```
= 0     0     [8] Travel value per pulse of encoder (mdeg/pulse)
= 0     0     [9] Rotation center position X of arc conveyor (mm)
= 0     0     [10] Rotation center position Y of arc conveyor (mm)
= 0     0     [11] Dummy data
= 0     0     [12] Dummy data
= 0     0     [13] Dummy data
= 0     0     [14] Dummy data
= 0     0     [15] Dummy data
```

3.1 [1] Encoder Rotating Direction

Data format : Integer
Data unit :
Data setting range : 0 or 1
Example of value : 0

When the startup order of phase A and phase B is compared at pulse output of phase A and phase B as output from the encoder used, the pulse adding direction is specified based on phase A. The value of this parameter is decided according to the encoder specifications used and the rotating direction of the encoder set to the conveyor.

If the "CONV" menu is chosen after pushing the "Utility" key, the following menu screens will appear.



Please work a conveyer, and set up a parameter so that the display of "Pulse" increases among a figure.

(Example)

1). (Encoder rotation direction value=0) & ("Pulse" increases)
Please set "0".

2). (Encoder rotation direction value=0) & ("Pulse" decreases)
Please set “1”.

3.2 [2] Encoder Interpolation Sampling Period

Data format	:	Integer
Data unit	:	msec
Data setting range	:	3 ~ 30000
Example of value	:	3

This parameter specifies the period during which the counter value from the encoder attached to the conveyor is obtained.

* As the arithmetic operation period of the conveyor synchronization function is three (3) ms, the actual parameter value is a multiple of this arithmetic operation period (3 ms) whose fractions are ignored.

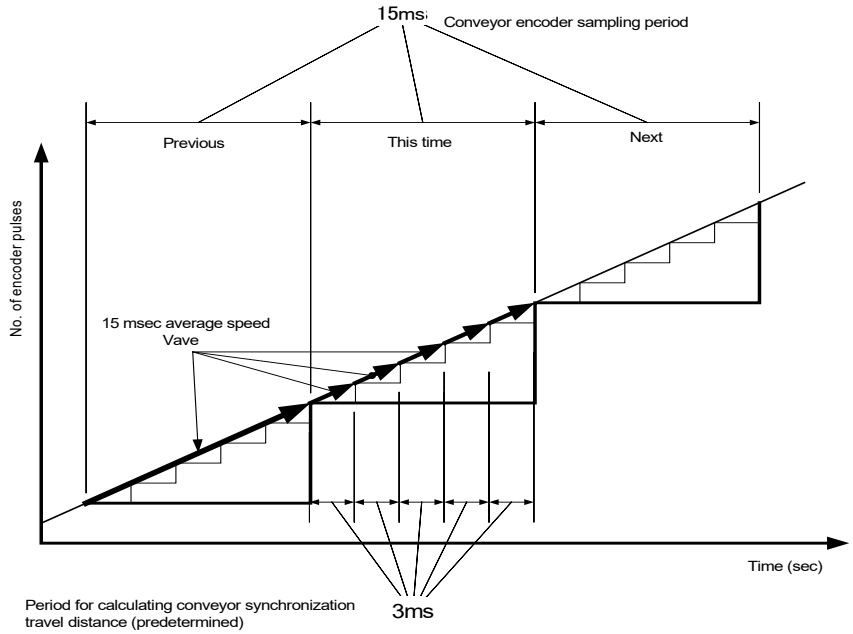
[Cautions]
The conveyor synchronization function calculates the synchronization travel distance at minimum periods of 3 ms (default). It is also possible to specify a larger value, however. Determine the value, taking note of the following.

- When the value decreases (minimum value 3 msec):
  - The trackability for the conveyor speed improves.
  - If this value is too small, however, the synchronization speed changes even with conveyor pulsation, which may result in vibration of the robot.

- When the value increases:
  - The synchronization speed is less affected by conveyor pulsation.
  - The conveyor rotating angle arithmetic error can be absorbed.
  - The trackability for the conveyor speed deteriorates.

Unless the pulsation is caused, specify the default value (3 msec).

Example:
When the conveyor sampling period is set to 15 msec, the synchronization travel distance is figured out based on the previous conveyor sampling period average speed (Vave).



3.3 [3] Conveyor Stop Judgment Time

This parameter specifies the conveyor stop judgment time. Unless the encoder counter value changes during the time set by this parameter, the system interprets that the conveyor is stopped and outputs the error of "001–048: Conveyor Stop". As this is the error of level 1, programmed operation continues.

Data format : Integer
Data unit : msec
Data setting range : 0 ~ 30000
Example of value : 3000

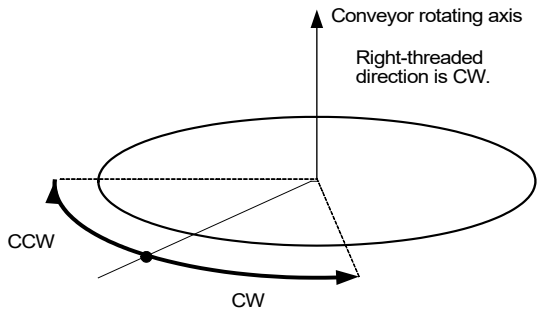
- Normally, the conveyor stop time is specified by the 2nd argument of SYNC command of the SCOL language. Unless the 2nd argument is specified, the conveyor stop is judged with this parameter value.
- When the 2nd argument of the SYNC command is not specified and this parameter value is zero (0), the system will not judge the conveyor stop.

3.4 [4] Type of Conveyor Synchronization

Data format : Integer
Data setting range : 0 ~ 2
Example of value : 0

This parameter specifies the type of the conveyor synchronization function. Select from the following three (3) types.

0 : Linear conveyor
1 : Arc conveyor (CW)
2 : Arc conveyor (CCW)



For the rotating direction of arc conveyor synchronization, the right-threaded direction of the conveyor rotating axis (*) is taken as CW.

* The conveyor rotating axis signifies the vector in parallel with the Z-axis of the world coordinate system, which passes the [8], [9] arc conveyor rotation center position.

3.5 [5] ~ [7] Dummy Data

Data format : Integer

Areas reserved for the system.

3.6 [8] Travel Value per Pulse of Encoder

This parameter specifies the conveyor travel value (mdeg) per one (1) pulse of the encoder set to the conveyor.

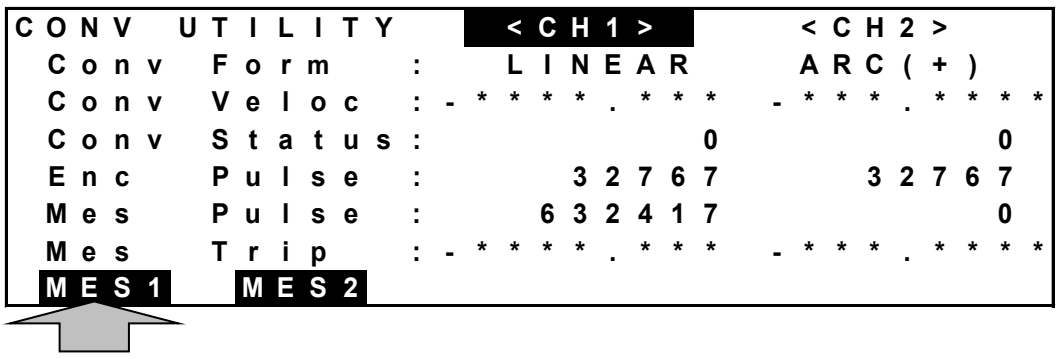
Data format : Real number
Data unit : mdeg
Example of value : 0

If the total number of encoder pulses is ΔP (pulse) when the conveyor moves by ΔL (deg), the conveyor travel distance per one (1) pulse of the encoder α (mdeg) can be calculated from the following equation.

$\alpha = (\Delta L / \Delta P) * 1000$

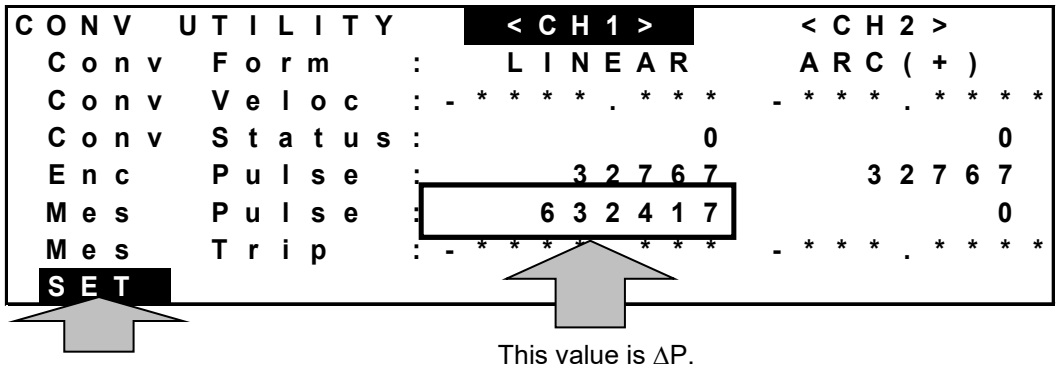
The total number of encoder pulses (ΔP) can be obtained in the following manner.

(1) Press the desired function key (<F1: MES1> or <F2: MES2>) for the channel (CH) you wish to measure, using the cursor.



When you wish to count the number of pulses of CH1, press <MES1>.

(2) To count the number of pulses of CH1 or CH2, select <MES1> or <MES2>, respectively. After moving the conveyor by ΔL, press the <SET> key.

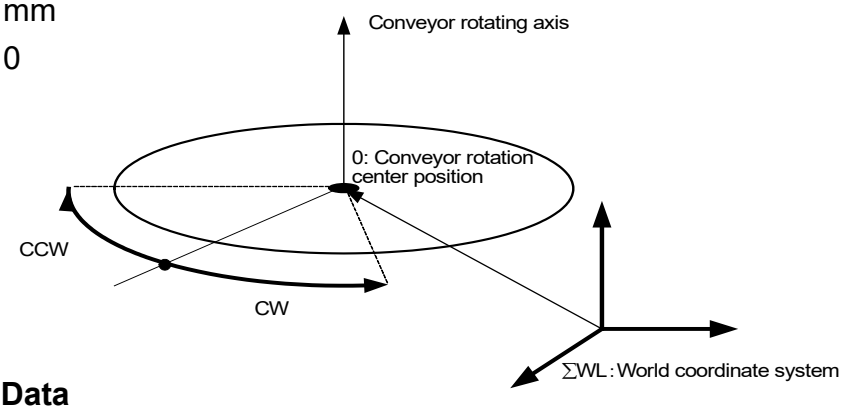


(3) When you press the <SET> key, the present total number of pulses (Mes Pulse) is shown on the display. Then figure out the travel distance per one (1) pulse (α μm) by substituting this value for "ΔP" in the above equation. (To measure the value again, press the <MES1> or <MES2> key.)

3.7 [9] [10] Rotation Center Position X, Y of Arc Conveyor (World Coordinate System)

These parameters specify the rotation center position X, Y (mm) of the arc conveyor when viewed from the origin of the world coordinate system.

Data format : Real number
Data unit : mm
Example of value : 0



3.8 [11] ~ [15] Dummy Data

Data format : Real number

Areas reserved for the system.

4. Commands

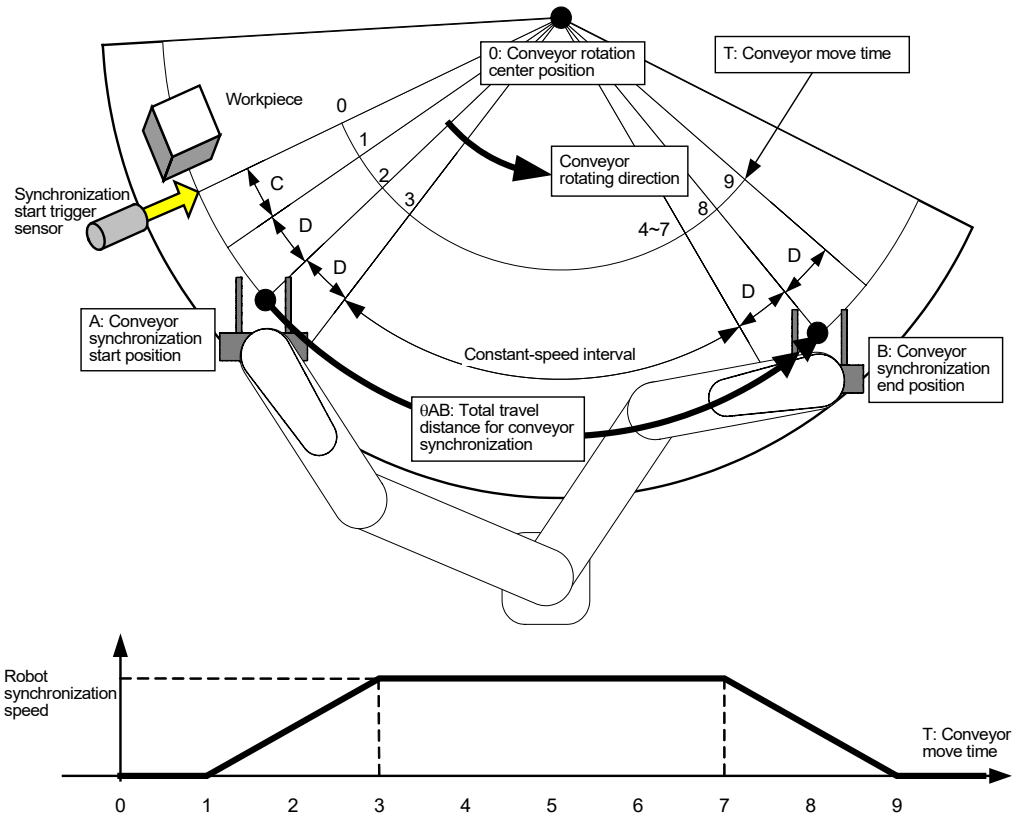
This section deals with the commands used for the conveyor synchronization function.

The following five (5) commands are newly added to the SCOL language (Symbolic Code Language for Robot) for the conveyor synchronization function.

Command	Function
CONV	Defines the conveyor synchronization parameters.
SYNC	Starts the conveyor synchronization.
UNSYNC	Finishes the conveyor synchronization.
XIN	Monitors the encoder cable breakage.
CNVSTAT	Accesses the information on conveyor synchronization status.

CONV

Function	:	Defines the conveyor synchronization parameters.
Format	:	CONV (<Number>, <Position 1>, <Position 2>, <Relative distance 1>, <Relative distance 2>)
Example	:	CONV (N, A, B, C, D)
Explanation & caution	:	Defines the parameters required for conveyor synchronization.



<Number; N> Logical conveyor number

Specify the logical number of the conveyor to be synchronized. Either 1 or 2 can be designated.
(A constant or variable of scalar type data can be used.)

<Position 1; A> Conveyor synchronization start position (position type data)

Specify the position for starting the conveyor and robot synchronization. The robot starts synchronization with the conveyor from this position. (Only X and Y components of position type data can be specified.)

<Position 2; B> Conveyor synchronization end position (position type data)

Specify the position for ending the conveyor and robot synchronization. The robot finishes synchronization with the conveyor at this position. (Only X and Y components of position type data can be specified.)

<Relative distance 1; C> Conveyor synchronization delay distance (deg)

Specify the relative angle between the synchronization start trigger signal ON point and the robot waiting point, considering a delay in the servo response, etc.
This parameter should be specified to compensate for the physical synchronization delay. If this value is too large, however, an error will occur. To avoid this, this value should be used for fine adjustment and the system should be configured so as to minimize this distance, where possible.

<Relative distance 2; D> Conveyor synchronization acceleration/deceleration distance (deg)

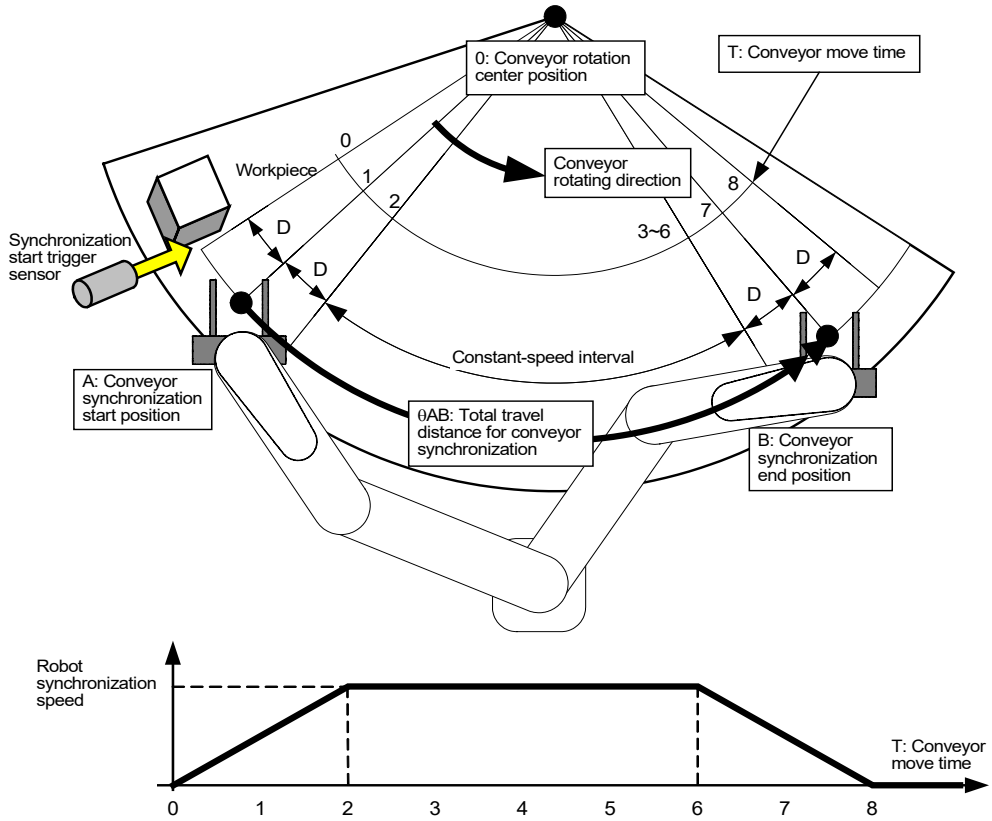
Specify the acceleration/deceleration distance (unit: deg) required for conveyor synchronization.
Unless the condition of $D \leq \frac{|\theta_{AB}|}{2}$ is satisfied, the error saying "002–033: Conveyor Parameter Error" generates and the robot stops.
When the conveyor has moved by the distance of "θAB" after the start of synchronization, the conveyor synchronization finishes.

Cautions

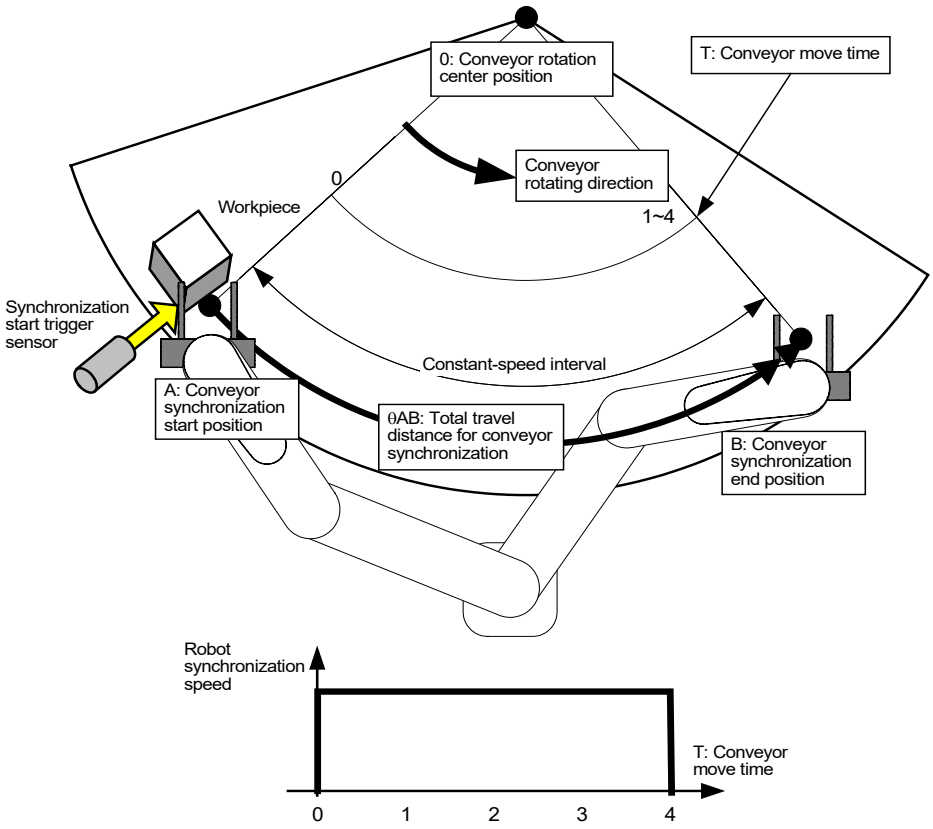
- Once defined, the conveyor synchronization parameter data are stored in the internal memory until the controller power is turned off. To change the conveyor synchronization parameter value, be sure to execute the new CONV command or perform the reset operation (*).

  * For the reset operation, see Section 7.

- Zero (0) can be specified for each of parameters C and D. When D (acceleration/deceleration distance) = 0, however, the synchronization speed of the robot is sharply increased to the conveyor speed, as shown below, which results in an excessively large load exerted on the robot.



When C = 0


0: Conveyor rotation center position
T: Conveyor move time
Conveyor rotating direction
0
1~4
Workpiece
Synchronization start trigger sensor
Constant-speed interval
A: Conveyor synchronization start position
θAB: Total travel distance for conveyor synchronization
B: Conveyor synchronization end position
Robot synchronization speed
T: Conveyor move time
0
1
2
3
4

When C = D = 0

SYNC

Function	:	Starts the conveyor synchronization.
Format	:	SYNC (<Number>, [<Stop judgment time>])
Example	:	SYNC (N, T)
Explanation & caution	:	At the same time that the synchronization start trigger signal is ON, the conveyor synchronization starts.

<Number; N> Specify the logical number of the conveyor to be synchronized.
Either 1 or 2 can be designated.
A constant or variable of scalar type data can be used.

<Stop judgment time; T>
Specify the conveyor stop judgment time. Unless this value is specified, the stop judgment is executed with the conveyor stop judgment time of user parameter [U20] (*).

* For the parameter setting method, see Section 3.

- Unless the conveyor synchronization parameter is defined by the CONV command, an error occurs when this command is executed.
- When starting the conveyor synchronization, be sure to describe the SYNC command at the beginning and the UNSYNC command at the end of the synchronization. Otherwise, an error occurs when this command is executed.
- Nesting is not possible for the SYNC command or UNSYNC command.
- If the stop operation is executed during waiting for the trigger signal ON, the conveyor synchronization will not start and the program stops. To restart the program, be sure to perform the reset operation, irrespective of the execution state (reset, retry or continue), then execute the program. For details, see Section 7.

UNSYNC

Function	:	Finishes the conveyor synchronization.
Format	:	UNSYNC (<Number>)
Example	:	UNSYNC (N)
Explanation & caution	:	The conveyor synchronization finishes.

<Number; N> Specify the logical number of the conveyor for which synchronization is to be finished.

- When a value other than zero (0) is specified for the acceleration/deceleration distance for conveyor synchronization [<Relative distance 2; D>, the robot decelerates and stops while the conveyor moves by "D" from the timing of this command execution.
- When the synchronization in conveyor synchronization travel distance "θAB" as explained in the paragraph of the CONV command has finished, the conveyor synchronization finishes automatically, irrespective of description of the UNSYNC command. (This does not mean that the UNSYNC command is unnecessary. Rather, the robot movement synchronized with the conveyor speed is stopped to assure the safety.)
- Unless this command is used together with the SYNC command described on the preceding page, an error occurs.

XIN

Function	:	Monitors the encoder cable breakage.
Format	:	XIN (<Signal name>)
Example	:	ON XIN (1) DO GOTO L100
Explanation & caution	:	Breakage of the encoder cable is monitored. If a fault is detected, it is informed.

<Number; N> Specify the logical number of the conveyor for which encoder cable breakage is to be monitored.

- This command is used to instantly detect a cable breakage. Use this command together with the ON command so that cable breakage can be detected at any time.
- When the ON–DO command is used together with this command, the number of simultaneous monitor counts should not exceed ten (10). If the value exceeds ten (10), an error occurs and the robot program stops accordingly.
- To cancel the monitoring of encoder cable breakage, use the IGNORE command.
  This command can be used together with the IF command and WAIT command.
- If the encoder cable has broken, the error saying "001–047 Conveyor Line Breakage" occurs. The robot will not stop, however, and continuously move, synchronized with the conveyor (encoder) speed effective just before the detection of cable breakage.
- The error of "001–047 Conveyor Line Breakage" is not reset as long as the power is turned off. When this error has occurred, turn the power off, identify and remedy the broken cable, then turn the power on again.

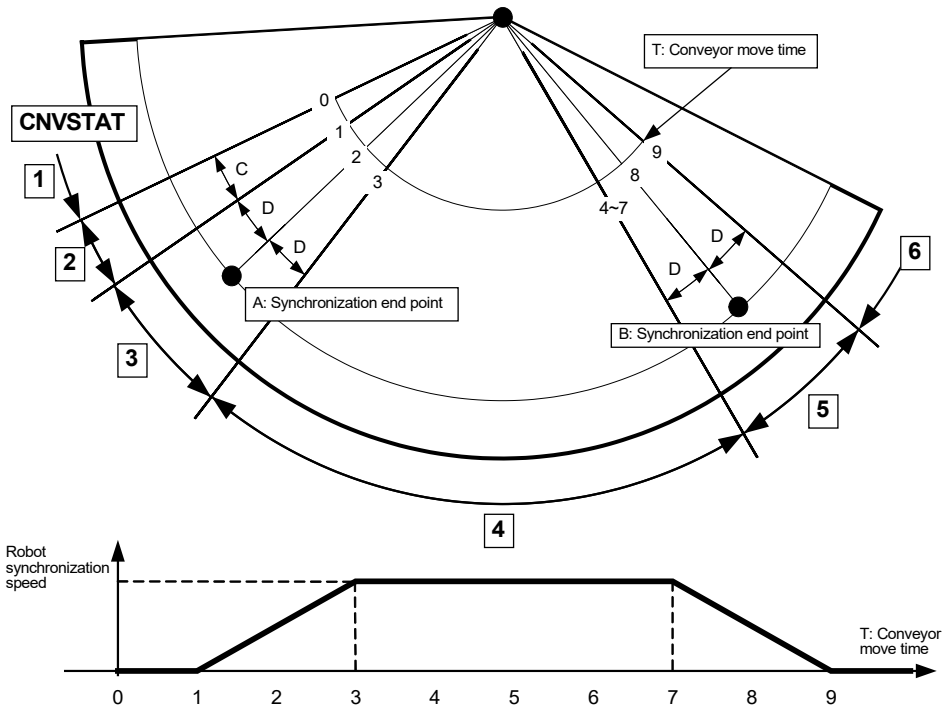
- If the encoder cable breakage has been detected during conveyor synchronization, be sure to recess the robot to the safe place. Otherwise, the robot and workpiece, or conveyor may collide. As you know from the sample program given below, recess the robot by means of the MOVES command after the detection of breakage, then finish the conveyor synchronization with the UNSYNC command. Finally, stop the program execution, using the STOP command.

Sample program

```
PROGRAM XINSAMPLE
    CONV (1, A, B, 40.0, 10.0)
    .....
L000:
    ON   XIN (1)   DO   GOTO   L100      Monitoring of encoder cable breakage before the start of
                                          synchronization.
    .....                                 (The monitoring should be cancelled just after the XIN
                                          command.  Breakage detection cannot be executed.)
    IGNORE   XIN (1)                      Cancel of monitoring before the start of synchronization.
    .....
    SYNC (1)
    ON   XIN (2)   DO   GOTO   L200      Monitoring of encoder cable breakage during
                                          synchronization.
    .....
    IGNORE   XIN (2)                      Cancel of monitoring during synchronization.
    UNSYNC (1)
    .....
    GOTO   L000
L100:
    PRINT   "C/V ERROR (1)", CR
    STOP
L200:
    MOVES   P1                            After the robot is recessed to the safe position with the
                                          linear interpolation control command, program execution
                                          stops.
    UNSYNC (N)
    PRINT   "C/V ERROR (2)", CR
    STOP
END
```

CNVSTAT

Function	:	Accesses the information on conveyor synchronization status.
Format	:	CNVSTAT
Example	:	WAIT CNVSTAT > = 4 ON COVSTAT < > 4 BREAK DO GOTO L1
Explanation & caution	:	The information on conveyor synchronization status is accessed.



Conveyor status table

0	Conveyor asynchronization	Signifies that the conveyor is not synchronized. (As this command can be used only after execution of the SYNC command, "1" is not set actually in the program.
1	Waiting for trigger signal	The conveyor is waiting until the trigger signal is turned on.
2	Conveyor moving in delay interval	The conveyor is moving in the delay interval.
3	Conveyor moving in acceleration interval	The conveyor is moving in the acceleration interval. During this time, the robot catches up with the conveyor speed.
4	Conveyor moving in constant-speed interval	The conveyor is moving in the constant-speed interval. The robot is moving by tracking the conveyor. Handling of a workpiece on the conveyor should be performed while CNVSTAT = 4.
5	Conveyor moving in deceleration interval	The conveyor is moving in the deceleration interval. During this time, the robot catches up with the conveyor speed.
6	Finish of conveyor synchronization	The conveyor synchronization has finished, and the conveyor is waiting for the UNSYNC command.

Cautions

- This command can be used during conveyor synchronization (i.e, after the SYNC command is executed).
- A value cannot be substituted for CNVSTAT.
- Only when the CNVSTAT conditional statement is used as the ON condition, the BREAK and PAUSE commands can be used.

Sample program

```
PROGRAM MAIN
    CONV (1, A, B, 100.0, 200.0)
    MOVE   P1
    SYNC (1)
    WAIT   CNVSTAT >=4
    MOVES   P2
    MOVES   P3
    WAUT   CNVSTAT >=6
    UNSYNC (1)
END
```

5. Interface

Robot controller TS2000 uses the conveyor synchronization by mounting option printed board X8HL. This printed board is connected to the option slot of the controller.

5.1 High-Speed Signal Input Specifications (Conveyor Synchronization Trigger Signal Specifications)

The conveyor synchronization function detects the trigger signal by using this high-speed signal input to assure more accurate detection of the conveyor encoder position while the trigger signal is ON.

The conveyor synchronization start trigger uses two (2) of the eight (8) high-speed inputs. Connect the CH1 conveyor trigger to high-speed signal input INP0 and the CH2 conveyor trigger to high-speed signal input INP1. (For details, see Para. 5.1.)

Also, when the high-speed input function is used for the conveyor synchronization start trigger switch, the remaining six (6) input switches other than the two (2) input switches become inoperative. (When using eight (8) high-speed inputs, the position data latch function cannot be used together with the conveyor synchronization function.)

The high-speed signal input specifications of the option printed board (X8HL) are shown below.

Table 5.1 High-speed signal input specifications

Item	Specifications
No. of inputs	Two (2) inputs (Of the total eight (8) inputs, only two (2) inputs are used for the conveyor synchronization.)
Input specification	Transistor open collector input or non-voltage contact input
Input signal (See Para. 5.1.1.)	CH1: INP0 CH2: INP1
Delay time	2.2 μsec (calculated value)

5.1.1 Circuit Structure

The circuit structures are shown below. As a bi-directional photocoupler is used in the input circuit, the source type (plus common) or the sink type (minus common) can be changed over.

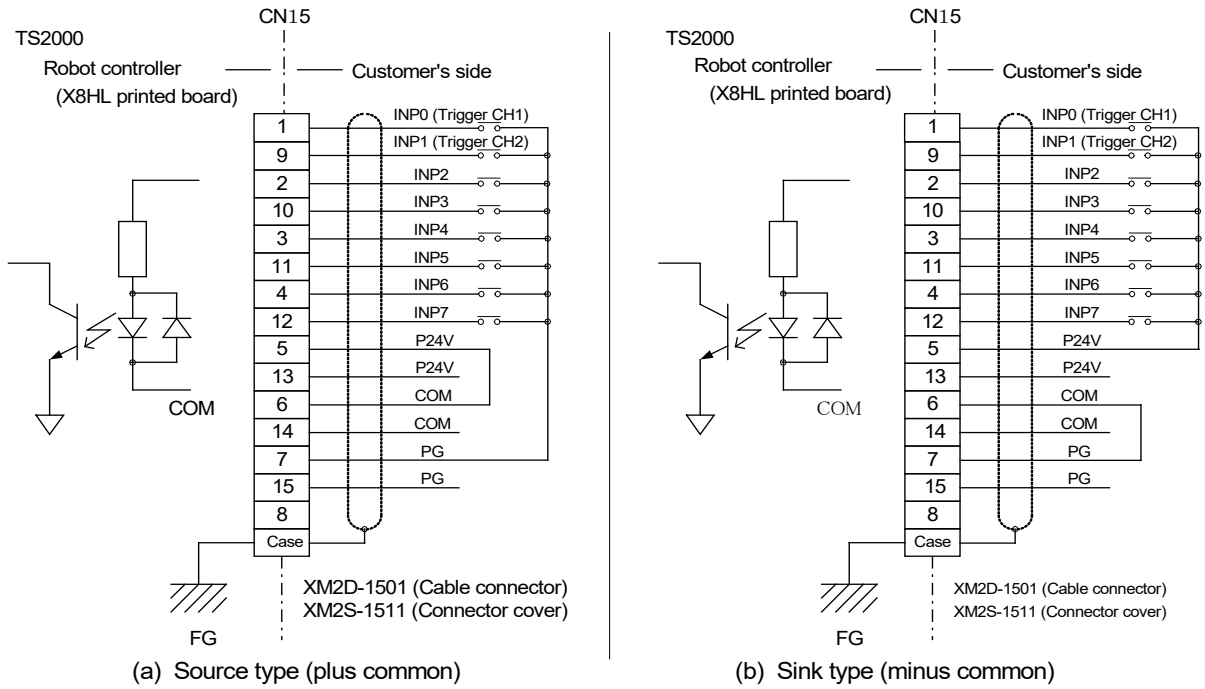


Fig. 5.1 Connection of high-speed input circuit

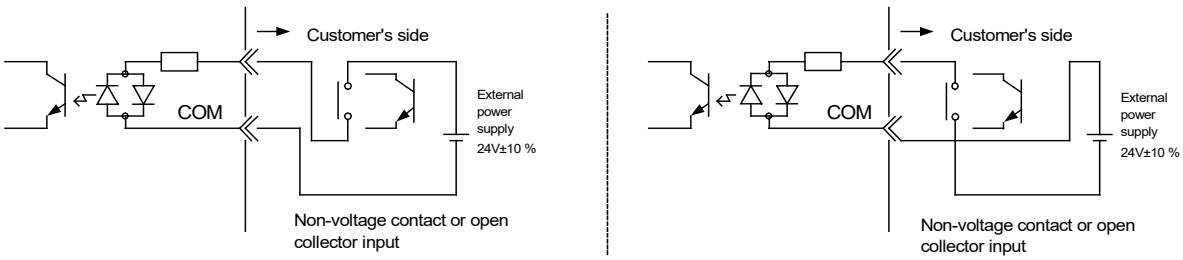


Fig. 5.2 Example of input signal circuit (when external power supply is used)

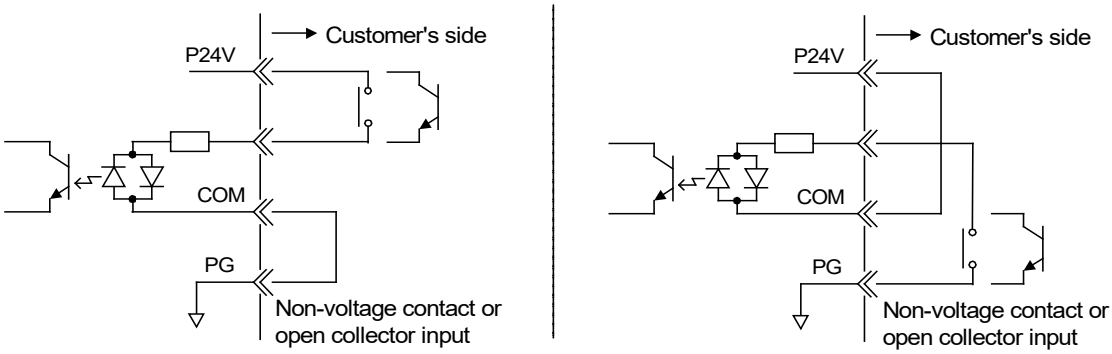


Fig. 5.3 Example of input signal circuit (when internal power supply is used)

5.1.2 Contact/Transistor Specifications

The contact/transistor connected with the external device should satisfy the following specifications.

Table 5.2 Non-voltage contact/transistor specifications

Non-voltage contact specifications	Transistor specifications
Contact rating: DC24 V, 10 mA or over Circuit current: Approx. 7 mA Minimum current: DC24 V, 10 mA Connected impedance: 100 Ω or less	Voltage between collector and emitter: 30 V or over Current between collector and emitter: 10 mA Circuit current: Approx. 7 mA Leak current between collector and emitter: 100 μA

5.1.3 Connection between Conveyor Synchronization Start Trigger Switches

Connect the conveyor synchronization start trigger switch to either of the high-speed input conveyor synchronization trigger switch 1 or 2. The connection is shown in Fig. 5.6 and Fig. 5.7.

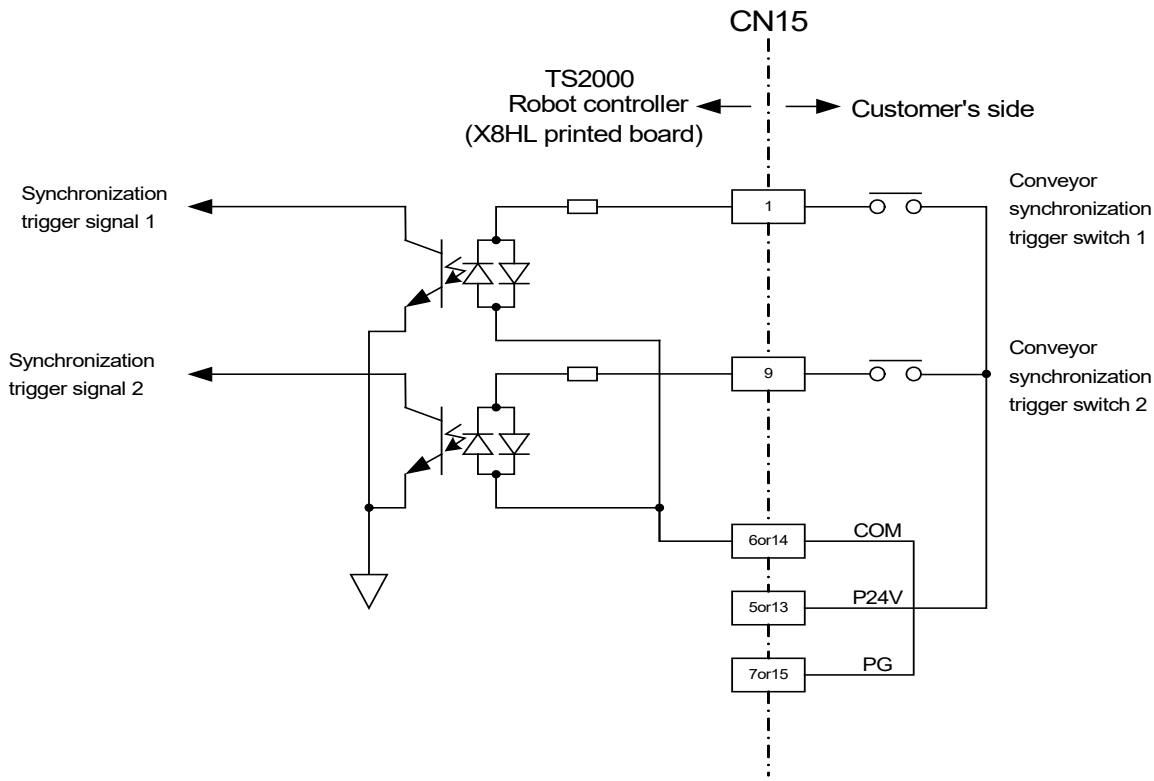


Fig. 5.6 Connection of conveyor synchronization start trigger switches (internal power supply; minus (–) common)

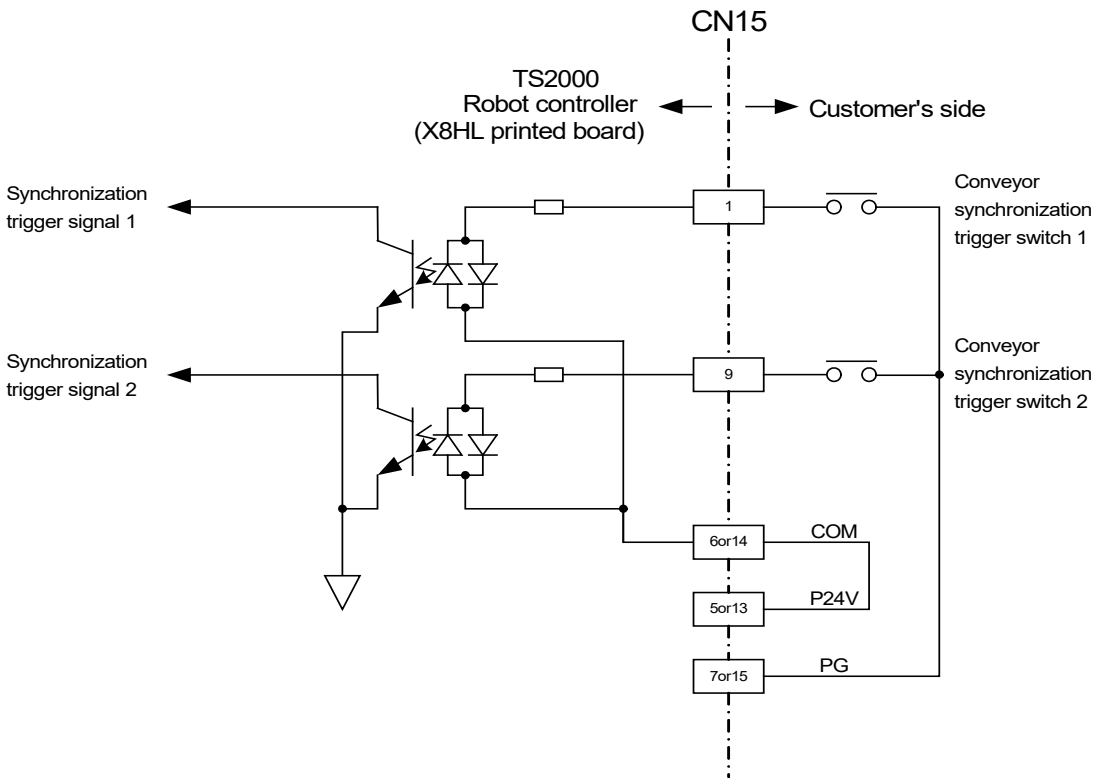


Fig. 5.7 Connection of conveyor synchronization start trigger switches (internal power supply; plus (+) common)

5.2 Conveyor Synchronization Specifications

When the conveyor synchronization function of the option printed board (X8HL) is used, the signal from the incremental encoder and input signal from the conveyor synchronization start trigger switch are required.

5.2.1 Incremental Encoder Specifications

Use such an encoder having phase A and B signals with 90° phase difference, whose output circuit is of an RS–422 differential output (line driver) type.

Table 5.3 Encoder specifications

Phases A/B maximum response frequency	250 kHz
Receiver IC	HD26LS32FP
Cable breakage detection	Provided

5.2.2 Incremental Encoder Connector Specifications

Connect the option printed board (X8HL) and incremental encoder to CN14 (*1).

*1: Connector CN14 on controller side: Type 52986–2071 made by Molex

Table 5.4 Connector and maker

Connector part name	Type and maker
Connector	54306–2011 (Molex)
Connector cover	54331–0200 (Molex)

For the connector signal array, see Para. 5.2.1.

5.2.3 Connection between Incremental Encoders

Use and connect a shield twisted pair cable as shown in Fig. 5.4.5. If it is connected incorrectly, noise resistance deteriorates, causing a malfunction. When the robot controller external power supply is used, connect according to Fig. 5.4. When the internal power supply is used, see Fig. 5.5. (The encoder input comes in two (2) channels. Use them when necessary.)

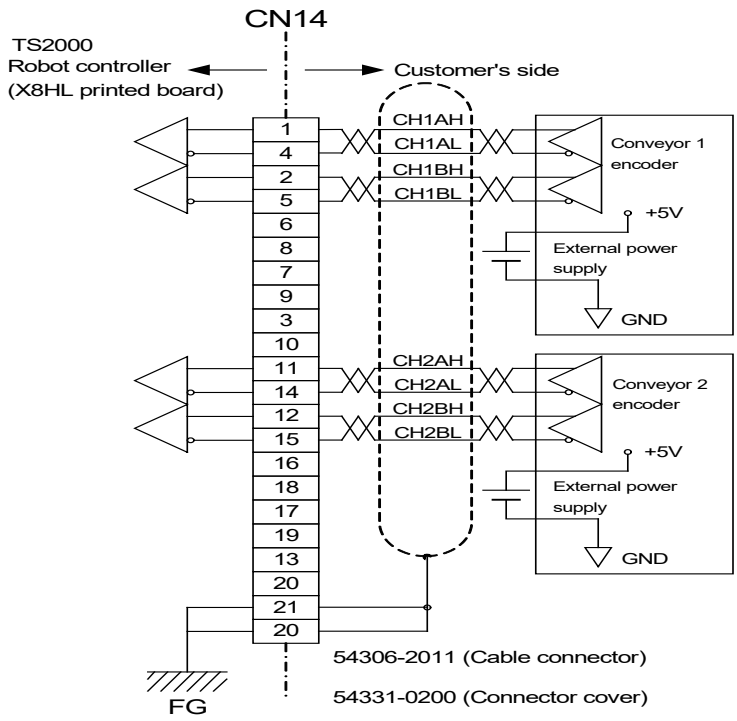


Fig. 5.4 Connection between incremental encoders (when external power supply is used)

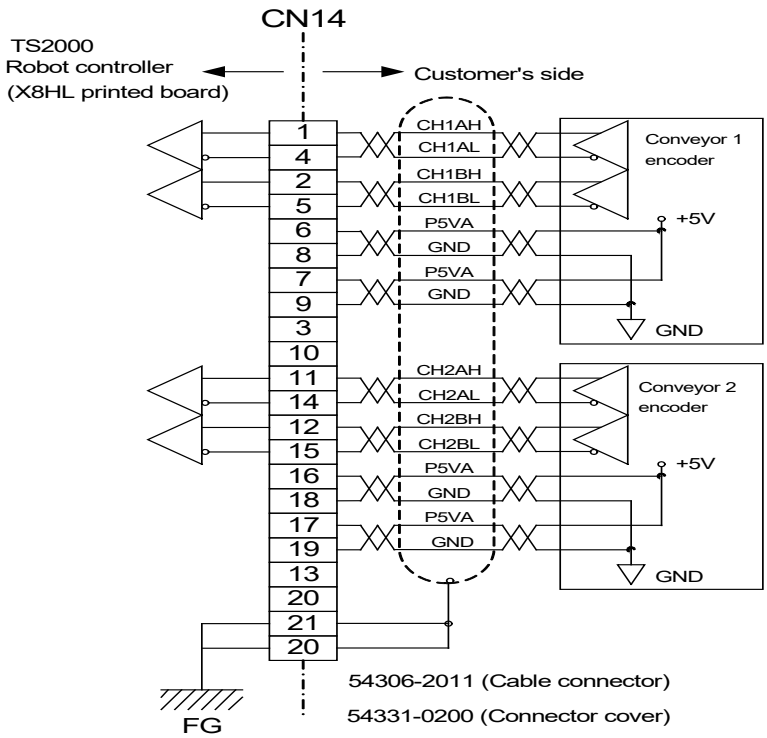


Fig. 5.5 Connection between incremental encoders (when internal power supply is used)

5.2.4 Relationship between Encoder Cable Length and Number of Power Supply Twisted Pairs

When +5 V supplied from the robot controller (i.e., option printed board) is used for the encoder power supply, connect the cable, considering voltage drop in the cable.

[Example of calculating cable voltage drop]

When the allowable encoder voltage is taken as 4.5 V, relationship between the cable length and number of power supply twisted pairs is as shown below.
(Assume that cable resistance is 110 Ω/km, encoder current consumption 150 mA and controller supply voltage 5 V.)

$$Ve = 5 - \frac{110}{1000} \times l \times 2 \times \frac{150}{1000} \times \frac{1}{p}$$

"Ve" indicates the encoder voltage, "*l* " the cable length, "2" the counts of reciprocation and "p" the number of twisted pairs.

From the above expression, it can be known that the applicable encoder voltage of 4.5 V can be assured under the following conditions.

No. of power supply twisted pairs	Cable length
1	15 m or less
2	30 m or less
3	45 m or less

When the cable of more than 45 m is required, therefore, the external power supply should be used.

The example of the above calculation is valid under the above conditions only. Figure out, referring to the example of the above calculation. Especially, note that the current consumption differs with an encoder to be used.
If the external power supply is used, the cable length can be longer. The maximum cable length should be 100 m, however, in terms of the signal.

5.3 Fabricating Cable

Fabricate a cable in the following manner.

a) High-speed signal input cable specifications

- Core cable: 0.2 mm^2 ~ 0.52 mm^2, twisted wires
- Cable outer diameter: Max. 9 mm
- Shield: Shield cluster
- Cable length: 30 m or less

b) Incremental encoder cable specifications

- Core cable: 0.2 mm^2 ~ 0.52 mm^2, twisted wires
- Cable outer diameter: Max. 8.5 mm
- Shield: Shield cluster
- Cable length: See Para. 5.2.4.

Use such a cable which satisfies the above specifications. Otherwise, the robot may malfunction due to noise entry, in addition to fusing caused by short-circuit or overheating of the core cable.

c) Fabricating a cable
The connector is of a solder cup type which solders wires.
To fabricate a cable, take careful precautions on soldering of the wires to the connector and shield processing, and refer to Fig. 5.8.

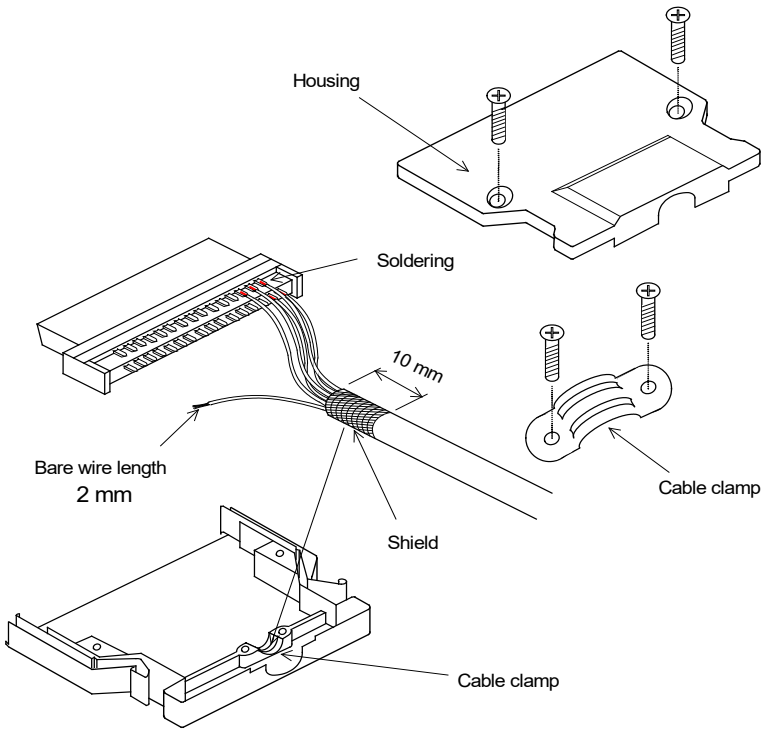


Fig. 5.8 Fabricating a cable

Note: The shield of the cable is connected with the grounding terminal of the TS2000 robot controller through the housing. The shield should be connected with the housing completely in the cable clamping part.

5.4 Assembling Printed Board

To assemble the option printed board (X8HL) in the robot controller, observe the following steps.

a) Turn off the power of the robot controller.

b) Cut off the 200 V power supplied from the robot controller. To assure safety, remove the power connector (ACIN CN1).

c) Remove the top cover of the robot controller.
The top cover is secured with nine (9) screws. That is, four (4) screws each are provided on both sides of the robot controller, and one (1) screw is provided on the top. Remove the screws, using a driver, then pull the cover toward you to remove.

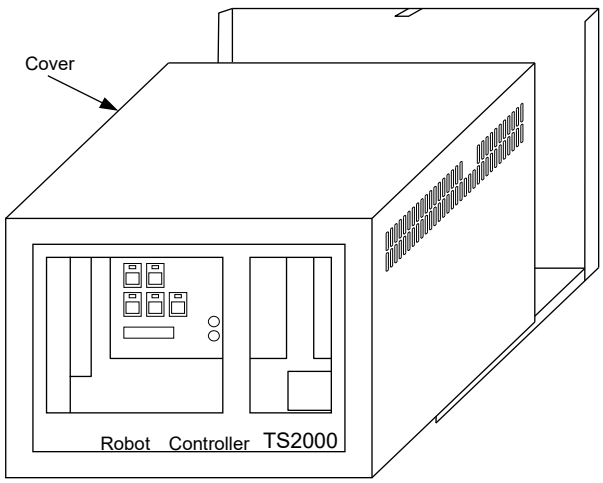


Fig. 5.9 Removing cover

d) Remove the dummy connector plate equipped on the rear side of the robot controller.

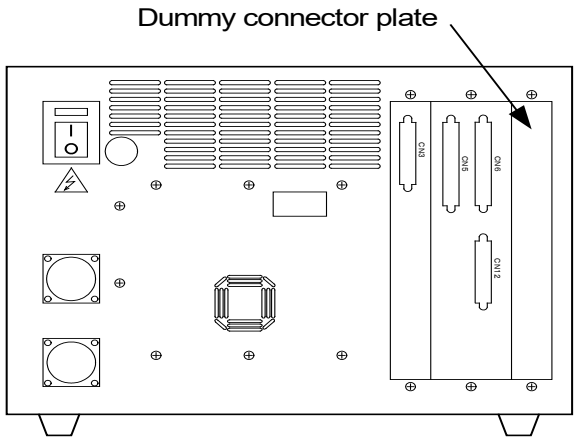


Fig. 5.10 Rear view of robot controller

e) Insert the option printed board (X8HL) into the mother board.
Remove the cover of connector CN6 equipped on the mother board (X8HB). Then insert the option printed board (X8HL) into CN6 of the X8HB printed board completely. After the insertion, secure the rear side of the robot controller (2 places) and inside steel platen (1 place) with M3 screws.

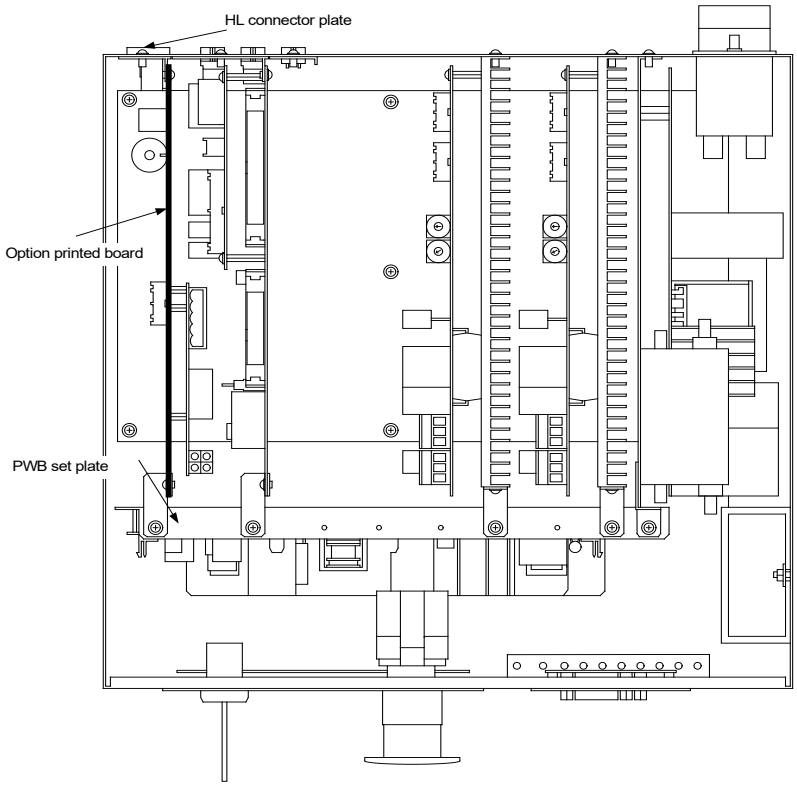


Fig. 5.11 Top view of robot controller

f) Cover the robot controller as it was before, then secure with the screws.

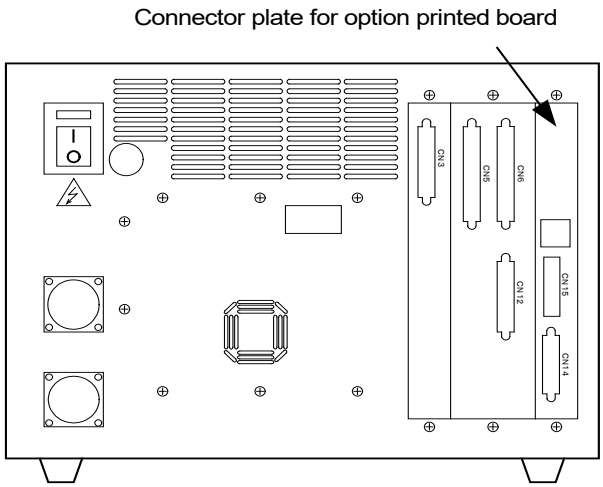


Fig. 5.12 Rear view of robot controller after assembly

5.5 Setting of terminator

When using the conveyor synchronization function, setting of the terminator is necessary because the encoder cable can be connected not only individually, but in a row as shown in Fig. 5.13 below and the conveyor position can be identified by each controller. The terminator setting is established by ON/OFF of SW4 (4 poles) equipped on the option (X8HL) printed board. A pair of SW4–1 and SW4–2 can be used for CH1. Likewise, a pair of SW4–3 and SW4–4 can serve for CH2. (See Table 5.1 below.)

All of four (4) poles of SW4 of the option (X8HL) printed board are factory-set to "ON" before shipment.

Table 5.1 SW4 setting (When the SW is set ON, the terminator is set.)

SW No.	Signal name	Setting	Remarks
SW4–1	CH1A	ON/OFF	Terminator setting switch for CH1 phase A data.
SW4–2	CH1B	ON/OFF	Terminator setting switch for CH1 phase B data.
SW4–3	CH2A	ON/OFF	Terminator setting switch for CH2 phase A data.
SW4–4	CH2B	ON/OFF	Terminator setting switch for CH2 phase B data.

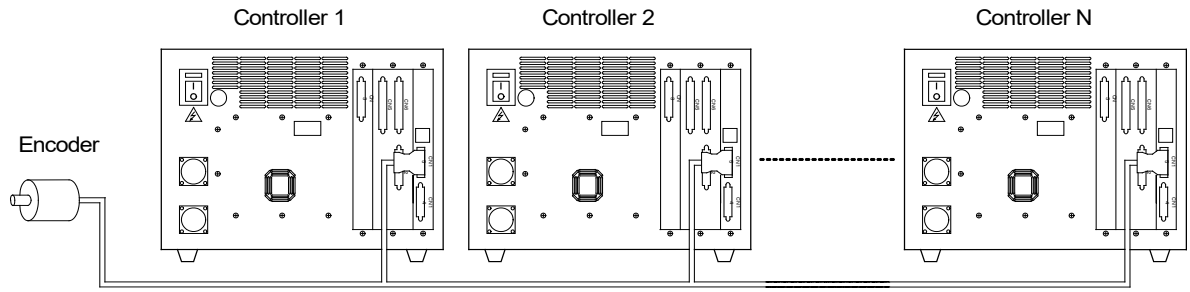


Fig. 5.13 Continuous connection of encoder

Set the SW4 in the following manner. When changing the setting, disconnect the X8HL printed board from the controller beforehand, referring to Steps a) through e) of Para. 5.4. (See Fig. 5.13 above.)

[1] When only one (1) controller (controller 1) is used:

a) When CH1 is used, set ON the SW4–1 and SW4–2.
b) When CH2 is used, set ON the SW4–3 and SW4–4.

* Even when only CH1 (or CH2) is used, it is recommended to set all four (4) poles of SW4.

[2] When two (2) controllers (controllers 1 and 2) are used:

a) Set OFF all four (4) poles of SW4 of controller 1.
b) Perform setting of SW4 of controller 2 in the same manner as in Para. [1] above.

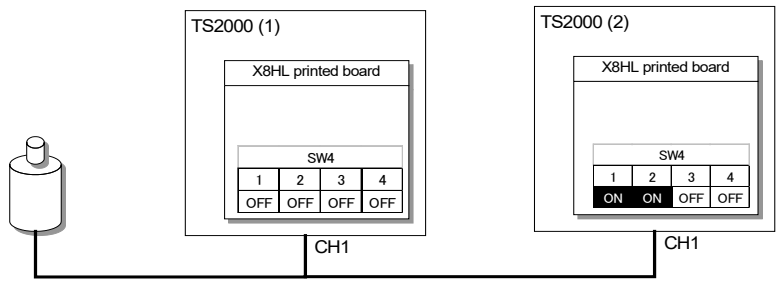


Fig. 5.14 Example of SW setting

[3] When three (3) or more controllers (controllers 1 ~ controller N) are used:

a) Perform setting of SW4 of controller N where the encoder cable is connected finally in the same manner as in Para. [1] above.
b) Set OFF all other SW4s (controller 1 ~ controller "N–1").

Note: Unless the above setting is performed, a precise value may not be displayed. To avoid this, be sure to perform and confirm the setting.

5.6 Defining Optimal Resolution of Encoder

The minimum processing period of conveyor synchronization is 3 msec. If the number of pulses of the conveyor encoder, that can be sampled every this period is too small, the tracking accuracy to the conveyor will deteriorate.
Select the encoder so that [V_{pulse}: Number of sampling pulses per 3 msec] as calculated from the following equation is about 10 ~ 100 pulses.

[V_{pulse}: Number of sampling pulses per 3 msec] can be figured out from the following equation.

$$V_{pulse} = \frac{3}{1000} \cdot \frac{1}{G_m} \left(\frac{V_{deg}}{360} \cdot R_e\right) \qquad (1)$$

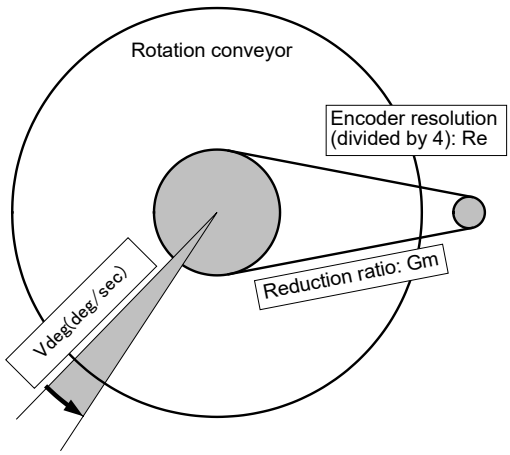
where,

Re: Encoder resolution (ppr)
When the 2000 (ppr; 1/4 of the value processed in the controller) resolution encoder is used, for instance, it is processed as 8000 (ppr) in the TS2000 controller.

V_{CNV}: Conveyor speed (deg/sec)
Customer's desired conveyor speed

V_{deg} = 10 (deg/sec)

Gm: Reduction ratio (Conveyor rotating speed / Encoder speed)
This value differs with the customer's encoder set conditions.



[Example] When the encoder revolves 160 turns while the conveyor rotates by one (1) turn:

G_m^* = 1/160

* The value of G_m (reduction ratio) should be exactly divisible.

From Expression (1) above, Expression (2) can be obtained.

$$R_e = V_{pulse} \cdot \frac{1000}{3} \cdot 360 \cdot \frac{G_m}{V_{deg}} \quad (2)$$

When the values are substituted for V_{deg}, G_m and V_{pulse} in Expression (2), the optimal encoder resolution (R_e) is about 750 ~ 7500 (ppr).

6. Programming and Teaching

6.1 Cautions on Programming

When creating the conveyor synchronization program, observe the following cautions.

(1) Motion commands that can be used

Motion control commands that can be described during processing control of SYNC command and UNSYN command (i.e., during conveyor synchronization) are only the linear interpolation (MOVES) and circular interpolation (MOVEC). If any other motion control command is programmed, the error of "002–025: Invalid command" occurs and the program stops with this command block. Once the synchronization has started, the robot continues moving in parallel with the conveyor coordinates until it reaches the conveyor synchronization end position (i.e., <Position 2; B> specified by CONV command).
This is also true for the motion control commands in the subprogram registered in the library file (SCOL. LIB).

(2) Setting of positioning accuracy: ACCUR = COARSE

When a motion control command is described during conveyor synchronization, specify "ACCUR = COARSE" for positioning accuracy of this motion control command. This should be programmed before the SYNC command. When "ACCUR = FINE" (initial state) is specified, conveyor synchronization is not executed for the motion control command.

(3) Robot configuration (Designation of CONFIG)

Be sure to specify the robot configuration effective during actual conveyor synchronization. That is, be sure to program "CONFIG = RIGHTY (right-shoulder system) or LEFTY (left-shoulder system)" before the SYNC command. If the robot configuration is illegal, the configuration currently kept internally is used compulsively by the SYNC command.

(4) ON ~ DO ~ Command (Conditional monitor)

- If monitoring of <Monitor condition> by ON command is required during conveyor synchronization, cancel it once by the IGNORE command just before the SYNC command, then specify it again.

- As in the XIN command, program "ON <Monitor condition> DO GOTO <Label>". Recess the robot when necessary and be sure to specify the conveyor synchronization end.

(5) PASS command (Shortcut motion)

When using a shortcut motion, specify "ENABLE PASS" just after the SYNC command and "DISABLE PASS" just before the UNSYNC command. If described with any other timing, the shortcut motion is interrupted or is not executed correctly.

(6) HERE command

When the HERE command is described during conveyor synchronization, this present position does not include the conveyor travel distance. Likewise, when the DEST command is used, the target position is unsettled. When using these commands, careful precautions should be taken.

(7) Distance for conveyor synchronization

When synchronization has finished in the total travel distance for conveyor synchronization "θAB" as described in the paragraph of the CONV command, the conveyor synchronization terminates automatically, irrespective of the presence or absence of the UNSYNC command.

If the program control command (WAIT ~ , etc.), input/output control command (INPUT ~, PRINT ~) or other command requiring some processing time is used, the conveyor synchronization continues for this duration and the robot reaches the conveyor synchronization end point. To avoid this, the synchronous work (workpiece chucking, etc.) should be programmed to complete in the total travel distance for conveyor synchronization "θAB".

(8) Program stop

When the STOP command is described during conveyor synchronization, the program stops, but the conveyor synchronization continues. This is also true as long as the conveyor (encoder) is working even after the program has stopped with other error. For details, see Section 7.

(9) Conveyor encoder cable breakage

- If the encoder cable has broken, the error saying "001–047 Conveyor Line Breakage" occurs. The robot will not stop, however, and continuously move, synchronized with the conveyor (encoder) speed effective just before the detection of cable breakage.

- If the encoder cable breakage has been detected during conveyor synchronization, be sure to recess the robot to the safe place. Otherwise, the robot and workpiece, or conveyor may collide. Recess the robot by means of the MOVES command after the detection of breakage, then finish the conveyor synchronization with the UNSYNC command. Finally, stop the program execution, using the STOP command.

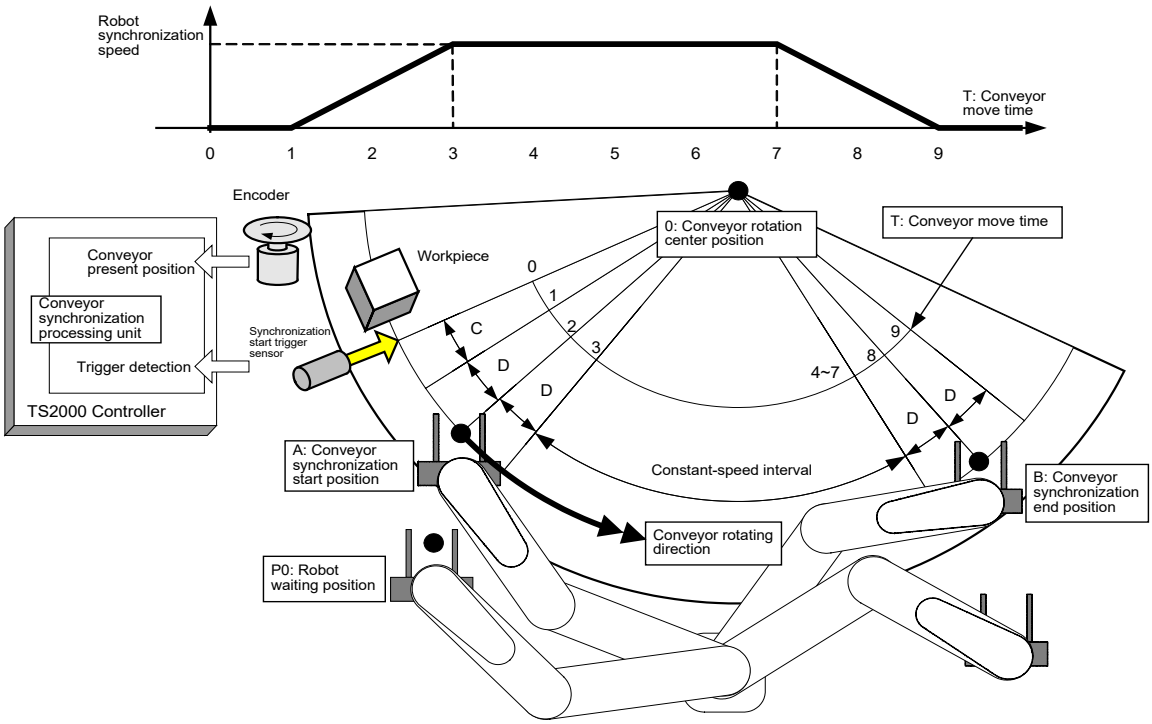
- The error of "001–047 Conveyor Line Breakage" is not reset as long as the power is turned off. When this error has occurred, turn the power off, identify and remedy the broken cable, then turn the power on again.

- Describe the XIN command just after the SYNC command and the IGNORE command just before the UNSYNC command. Even if the BREAK or PAUSE command is specified during conveyor synchronization, it is neglected. Program "ON <Monitor condition> DO GOTO <Label>."

6.2 Outline of Programming

This paragraph describes the conveyor synchronization programming by creating the basic program of conveyor synchronization.

[1] Specify synchronization parameters A, B, C and D of the CONV command.
[2] Before starting conveyor synchronization, the robot should be retracted to the conveyor synchronization waiting position/retract position (P0).
[3] At the same time that the trigger signal has been input, the SYNC command for starting the processing of conveyor synchronization should be defined.
[4] Specify the motion commands for the job on the conveyor.
[5] To finish the processing of conveyor synchronization, define the UNSYNC command.
[6] Select the program and make sure that no program format error is generated.



<Example program>

PROGRAM CONVTEST

	N=1	
	C=10.0	
	D=5.0	
	ACCUR=COARSE	Coarse positioning accuracy is selected.
	CONFIG=RIGHTY	The right-shoulder system is selected for robot configuration.
	CONV (N,A,B,C,D)	The parameters are defined.
	MOVE P0	The robot moves to the waiting position.
	SYNC (N)	Waiting for LS ON. With LS ON, synchronization starts.
	WAIT CONVSTAT >= 4	Waiting until the robot enters the constant-speed interval of conveyor synchronization.
Program of actual job synchronized with conveyor	MOVES P1+POINT (0,0,100)	The robot moves to the approach point.
	DOUT (213, –214)	The robot hand opens.
	MOVES P1 WITH SPEED = 20	The robot moves to the chucking point.
	DOUT (–213, 214)	The robot hand closes.
	DELAY 0.5	The robot motion stops for 0.5 sec.
	MOVES P1+POINT (0,0,100)	The hand moves up.
	MOVES P0	The robot moves to the retract point.
	UNSYNC (N)	Synchronization ends.

END

DATA

A = ~ Conveyor synchronization start point

B = ~ Conveyor synchronization end point

P0 = ~ Waiting point/retract point

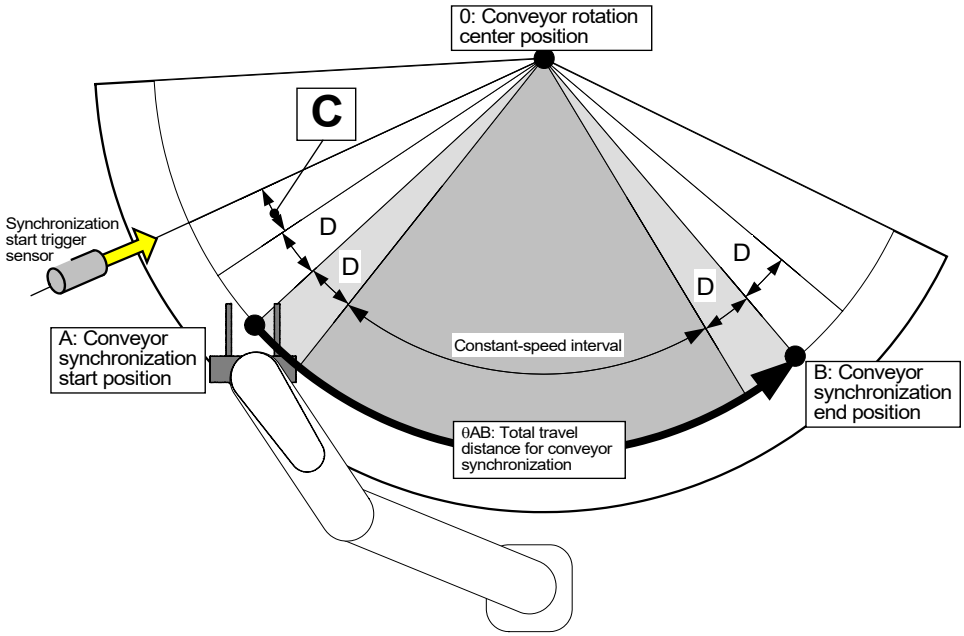
P1 = ~ Chucking point

END

6.3 Setting of CONV Command Parameters A, B, C, D

This paragraph describes the setting and adjustment of CONV command parameters A, B, C, D as described in the program.

6.3.1 Deciding [C: Conveyor Synchronization Delay Distance (deg)]



This parameter specifies the angle (deg) between the actual trigger sensor position and the conveyor synchronization start position indicated by point A (when [D: Conveyor acceleration/deceleration interval] = 0).

This parameter is useful when the trigger sensor should be located at a position distant from the conveyor synchronization start position due to the line layout. If this value is too large, however, an error will occur. To avoid this, the system should be configured so as to minimize this distance, where possible.

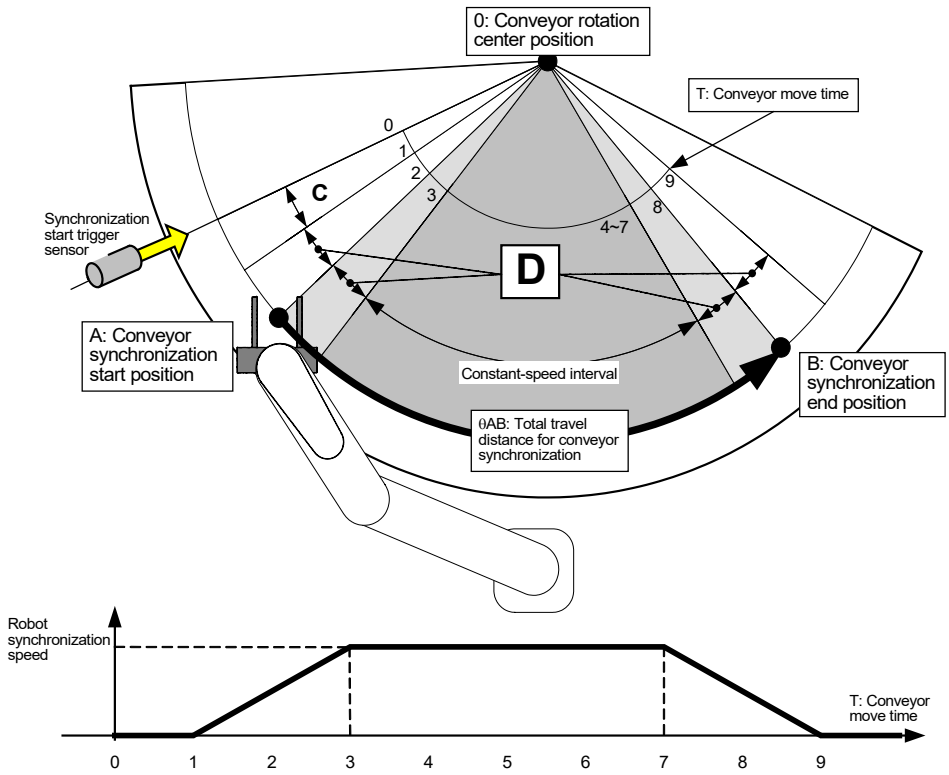
Additionally, if tracking delay (*1) during conveyor synchronization is caused, this parameter value should be specified between the synchronization start trigger sensor and robot acceleration start point to compensate it.

A negative value cannot be specified for [C: Conveyor synchronization delay interval]. If specified, the error saying "002–033: Conveyor Parameter Error" generates.

*1: The tracking delay originates from (1) response delay of servo control, (2) trigger signal input detection delay of the robot controller, etc.

6.3.2 Deciding [D: Conveyor Synch. Acceleration/Deceleration Interval (deg)]

Acceleration/deceleration interval D is intended for gradually accelerating the robot speed to the conveyor speed. Thus, an impact exerted on the robot or workpiece at the start of synchronization can be alleviated. It should be specified after interval C, as shown below. (This parameter cannot be specified before interval C.)



[1] The robot accelerates when [T: Conveyor move time] = 1 ~ 3. While the robot moves by "D" from [A: Synchronization start point], the conveyor moves "2 × D".

[2] At the end of conveyor synchronization, the conveyor decelerates when [T: Conveyor move time] = 7 ~ 9. While the robot enters the deceleration interval and moves by "D" to [B: Synchronization end point], the conveyor moves "2 × D".

[3] A negative value cannot be specified for [D: Conveyor synchronization acceleration/deceleration interval]. If specified, the error saying "002–033: Conveyor Parameter Error" generates.

A value (minimum value) providing a yardstick in determining the D value can be obtained from [A_{max} (mm/sec^2): Maximum acceleration for robot interpolation], [V: Conveyor synchronization speed (deg/sec) and [R: Rotating radius for conveyor synchronization].

Type of robot	Max. acceleration: A_{max} (mm/sec^2)
SR–654HSP/ SR–554HSP SR–854HSP/ SR–1054HSP SR–1504HSP	6,700
SR–424HSP	15,000

<Expression>

$$D_{min} << (V * R)^2 / 2 \bullet A_{max}$$

[V: Conveyor synchronization speed] is shown on the conveyor speed display [Veloc] of the conveyor utility function (*1). This is the minimum value, however, and should be increased according to the mass of the robot hand and workpiece chucked. Use a value multiplied by 2 ~ 3 at least.

$$D = 3 \bullet D_{min}$$

If there is an impact during acceleration to the conveyor speed at the stage of adjustment as described later, readjust the D value.

*1: For the conveyor utility function, see Para. 7.4.

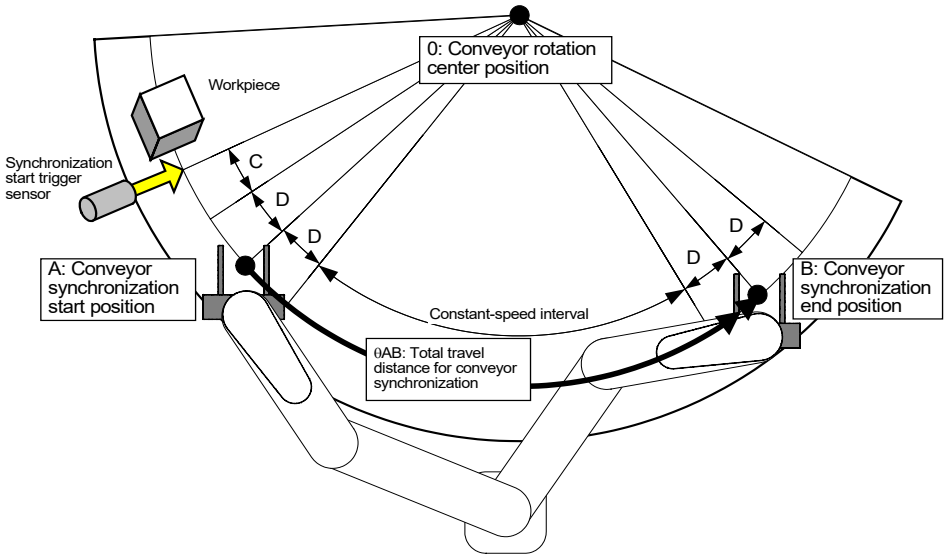
The actual job interval during conveyor synchronization is the constant-speed interval in which the robot synchronization speed coincides with the conveyor speed, and the distance of the constant-speed interval is figured out from the following expression.

Actual job interval = Travel distance in constant-speed interval

$$= \theta AB - 2 \times D$$

If the C or D value is larger than required, the actual robot job interval reduces accordingly. Specify an appropriate value.

6.3.3 Teaching [A: Conveyor Synchronization Start Position / B: Conveyor Synchronization End Position]



[Total travel distance for conveyor synchronization: θAB] includes the acceleration/ deceleration interval (D) required for conveyor synchronization, too. However, the interval in which the robot is completely tracking the conveyor speed is the constant-speed interval.

Teach point A so that it can be located at a position advanced by the distance of "C+D" in the conveyor vector direction from the trigger sensor position. (*1)

*1: A teaching error caused here in the conveyor vector direction can be absorbed by adjusting the value of [C: Conveyor synchronization delay distance].

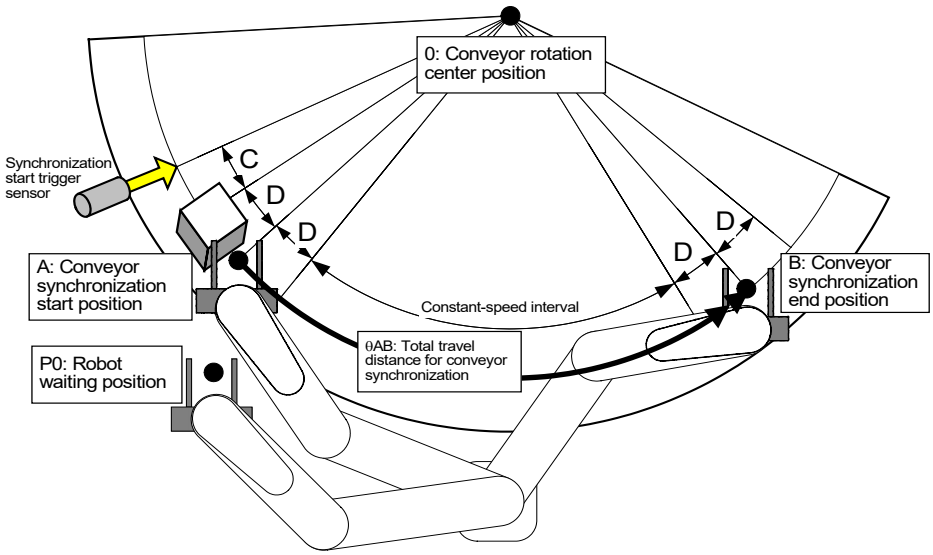
6.4 Teaching

This paragraph deals with the teaching operation of the conveyor synchronization function.

! CAUTION

- Before teaching operation, be sure to stop the conveyor. If the conveyor is moving, the synchronization start trigger sensor may operate under some teaching condition, resulting in an unexpected movement or collision. (The robot at the robot waiting position is only kept stationary until the trigger sensor turns on, and it can start moving at any time.)
- The robot retract position should be the completely safe position where it will not collide with a workpiece running on the conveyor even if the robot operation is stopped. (Desirably, the robot waiting position should serve as the robot retract position.)
- The trigger sensor should be adjusted so that it can always be turned on at a predetermined position without causing shift of the workpiece detecting position. If the sensor is turned on due to mixture of different workpieces or at a position other than the adjusted position, the robot tracking shift or collision may be caused.
- While the test operation mode is selected, DO NOT execute the step of the program (between SYNC command to UNSYNC command), machine lock (dry mode operation) or [MODE: Motion]. If the conveyor is running, an expected collision may be caused.

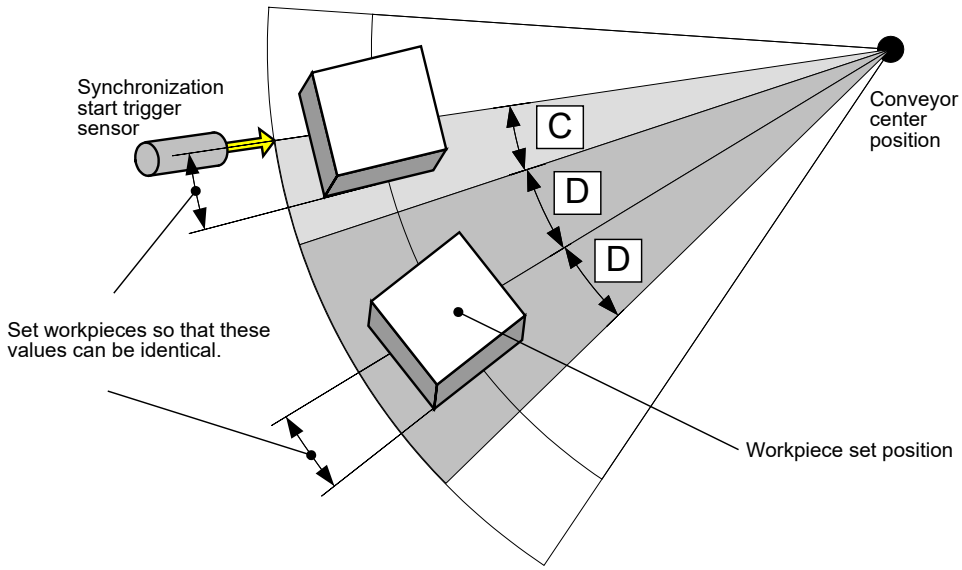
6.4.1 Teaching of Conveyor Waiting Position



Teach the robot waiting position (P0).

6.4.2 Teaching of Motion Points during Conveyor Synchronization

Teach the motion points used in the motion command during conveyor synchronization. When teaching the robot actually, use a workpiece placed on the conveyor. At this time, take care of the relationship between the trigger sensor set position and workpiece detecting position.



When the total of the conveyor speed (converted from deg/sec to mm/sec) and motion speed of motion command exceeds 1000 (mm/sec), the error saying "Axis1 ~ 4 Exceed Speed" occurs and the robot may stop. When using a motion command in the same direction of conveyor rotation, take careful precautions on the programming of the motion speed.

6.5 Readjustment and Confirmation of Conveyor Synchronization Parameters [CONV (A, B, C, D)]

Confirm the conveyor synchronization parameters. In this paragraph, check is performed on whether the actual job program during conveyor synchronization can be finished in the constant-speed interval of the conveyor synchronization. For this purpose, the following condition should be satisfied.

$\theta AB - 2 \times D >> \theta cv$ (deg): Conveyor travel distance during actual job

Unless the synchronization can complete, make adjustment in the following manner.

(A) Increase "θAB".
(B) Minimize the travel distance in the C and D intervals.
(C) Reduce the processing time of the actual job program.
(D) Reduce the conveyor speed.
(E) Change the trigger sensor position.

Step 1: Measure the processing time (Tcv (sec) of the actual job program during conveyor synchronization between the SYNC command and UNSYNC command.

```
PROGRAM CONVTEST
    :
    CONV( N, A, B, C, D )
    MOVE P0
    ' SYNC(N)              →  Comment out
    TIMER = 100.0
    'WAIT CONVSTAT >= 4    →  Comment out
    MOVES P1 + POINT( 0, 0, 100 )
    DOUT(213,–214)
    MOVES P1 WITH SPEED = 20
    DOUT(–213, 214)
    DELAY 0.3
    MOVES P1+POINT( 0, 0, 100 )
    MOVES P0
    PRINT "Tcv=", TIMER–100.0, CR  →  The processing time (sec) is displayed on the teach pendant.
    ' UNSYNC(N)           →  Comment out
END
```

Actual job program during conveyor synchronization
Tcv (sec): Actual job time
Lcv (mm): Conveyor travel distance during actual job

Cautions:

1. As this program uses the conveyor synchronization start command as the comment out, the conveyor synchronization is not executed.

2. Before executing this program, remove a workpiece left on the conveyor and stop the conveyor.

Step 2: Confirm the motions of this program by the test operation (MODE: MANU), execute the program in the internal automatic operation (MODE: INT) mode, then measure the processing time. The processing time is displayed under [Tcv (sec)] on the teach pendant after the end of program execution.

Step 3: Retract the robot to the safe position and start the conveyor.
Measure the conveyor speed [Vcv (deg/sec)], using the conveyor utility function (*1). When the values of Tcv and Vcv are substituted for the following expression, the conveyor travel distance during actual job [θcv (deg)] can be obtained. If this distance is small enough, compared with the distance of constant-speed interval (or if the following condition is satisfied), setting and confirmation of the CONV (A, B, C, D) parameters complete.

$$|\overrightarrow{AB}| - 2 \cdot D >> \theta cv \text{ (mm)} = Vcv \times Tcv$$

*1: For how to use the conveyor utility function, see Para. 7.4.

Step 4: Confirm the trackability for conveyor synchronization.
Actually feed workpieces on the conveyor, and when the trigger signal has turned on, the conveyor synchronization starts. To make the conveyor synchronization effective, however, observe the following operation.

(A) Remove the workpieces.
(B) Make the SYNC and UNSYNC command valid.
(C) Set all actual job programs as the comment out.

(D) If the waiting position differs from [A: Synchronization start position] and the distance between the conveyor and robot increases, check of the trackability is rather difficult. When this happens, change the waiting position (P0) to [A: Synchronization start position].

```
PROGRAM CONVTEST
    :
    CONV( N, A, B, C, D )
    MOVE P0
    MOVE   A              →    Change the waiting position to point A.
    SYNC(N)
    WAIT CONVSTAT >= 4
     ‘MOVES P1 + POINT( 0, 0, 100 )
    ‘DOUT(213,–214)
    ‘MOVES P1 WITH SPEED = 20
    ‘DOUT(–213, 214)
    ‘DELAY 0.3
    ‘MOVES P1+POINT( 0, 0, 100 )
    ‘MOVES P0
    UNSYNC(N)
END
```

Actual job program during conveyor synchronization
→ Set all lines (or blocks) as comment out.

Repeat the synchronized movement, using this program and perform check on the following items.

a. Make sure that no impact is caused to the arm at the start of conveyor synchronization.
Adjust [D: Conveyor acceleration/deceleration interval] according to Para. 6.3.2.

b. Make sure of the robot waiting position.
If the robot is behind the workpiece just after the start of synchronization, decrease the C value.
If the robot is ahead of the workpiece just after the start of synchronization, increase the C value.

c. Make sure of the travel distance per one (1) pulse of encoder (*2) of [USER. PAR].
If the robot lags behind the workpiece in the latter half of the conveyor synchronization distance, increase the travel distance per one (1) pulse of encoder.

If the robot is faster in the latter half of the conveyor synchronization distance, decrease the travel distance per one (1) pulse of encoder.

*2: See Para. 3.6.

Now, readjustment and confirmation of conveyor synchronization parameters CONV (A, B, C, D) have finished.

6.6 Confirming Programmed Operation

After confirming that the trackability is sufficient for the workpiece, carefully return the program to the original state, then perform actual job such as workpiece chucking.

! CAUTION

- Before starting the program, read through Section 7. Especially, after the stop operation during conveyor synchronization, observe the restoring procedures described in Para. 7.1 and 7.2.
- In the test operation mode, the speed limit (LSPEED) is limited to 25 %. Thus, the synchronous work (workpiece chucking, etc.) will not complete in the synchronized movement distance, which is very dangerous. DO NOT execute the test operation if the work to the preceding paragraph has completed.
- DO NOT change the value of LSPEED, but confirm it by the internal automatic operation.
- If test operation is required after this step, stop the conveyor and make sure that no obstacle is left within the robot working envelope.

Now, the operation has finished. If there is any problem, repeat the procedures, starting from the beginning of Section 6.

<Test program for final confirmation>

```
PROGRAM CONVTEST
  N=1
  C=10.0
  D=5.0
  ACCUR=COARSE                  Coarse positioning accuracy is selected.
  CONFIG=RIGHTY                 The right-shoulder system is selected for robot
                                configuration.
  CONV (N,A,B,C,D)              The parameters are defined.
  MOVE P0                       The robot moves to the waiting position.
  SYNC (N)                      Waiting for LS ON.  With LS ON, synchronization starts.
    WAIT CONVSTAT >= 4          Waiting until the robot enters the constant-speed interval of
                                conveyor synchronization.
    MOVES P1+POINT (0,0,100)    The robot moves to the approach point.
    DOUT (213, –214)            The robot hand opens.
    MOVES P1 WITH SPEED = 20    The robot moves to the chucking point.
    DOUT (–213, 214)            The robot hand closes.
    DELAY 0.5                   The robot motion stops for 0.5 sec.
    MOVES P1+POINT (0,0,100)    The hand moves up.
    MOVES P0                    The robot moves to the retract point.
  UNSYNC (N)                    Synchronization ends.
END
DATA
  A = ~                         Conveyor synchronization start point
  B = ~                         Conveyor synchronization end point
  P0 = ~                        Waiting point/retract point
  P1 = ~                        Chucking point
END
```

6.7 Sample Programs

The programs using the conveyor synchronization control command are exemplified below. When using these programs for actual robot operation, review the programs according to the robot working envelope and surrounding condition.

6.7.1 Handling Work

```
PROGRAM HANDLING
  ACCUR=COARSE                  Coarse positioning accuracy is selected.
  CONFIG=RIGHTY                 The right-shoulder system is selected for robot
                                configuration.
  N=1
  C=10.0
  D=5.0
  CONV (N,A,B,C,D)              The parameters are defined.
  MOVE P0                       The robot moves to the waiting position.
  SYNC (N)                      Waiting for LS ON. With LS ON, synchronization starts.
    MOVES P1+POINT (0,0,100)    The robot moves to the approach point.
    DOUT (213, –214)            The robot hand opens.
    MOVES P1 WITH SPEED = 20    The robot moves to the chucking point.
    DOUT (–213, 214)            The robot hand closes.
    DELAY 0.5                   The robot motion stops for 0.5 sec.
    MOVES P1+POINT (0,0,100)    The hand moves up.
    MOVES P0                    The robot moves to the retract point.
  UNSYNC (N)                    Synchronization ends.
END
DATA
  A = ~                         Conveyor synchronization start point
  B = ~                         Conveyor synchronization end point
  P0 = ~                        Waiting point/retract point
  P1 = ~                        Chucking point
END
```

- Be sure to specify the positioning accuracy and robot configuration.
- Use variables for each parameter.
- For the robot waiting point, see Section 1.
- The hand is opened and closed with auxiliary output signals. (Use of DOUT command)
- Lower the robot operating speed as the Z-axis (i.e., vertical axis) is used. (Use of WITH statement)
- The retract point at this time is shifted toward the conveyor advance direction by the length of synchronized movement from actual point P1.

6.7.2 Pick & Place Work

```
PROGRAM PICKUP
  ACCUR=COARSE                Coarse positioning accuracy is selected.
  N=2
  CONV (N,A2,B2,15.0,10.0)    Conveyor 2 parameters are defined.
  ON XIN(2) DO GOTO L200      Encoder cable 2 is checked for breakage.
  N=1
  CONV (N,A1,B1,40.0,10.0)    Conveyor 1 parameters are defined.
  ON XIN(1) DO GOTO L200      Encoder cable 1 is checked for breakage.
            :
L100:
  CONFIG=RIGHTY               The right-shoulder system is selected for robot
                              configuration.
  MOVE P1                     The robot moves to waiting point 1.
  SYNC (1)                    Waiting for LS1 ON.  With LS1 ON, synchronization starts.
            :                 Workpiece on conveyor 1 is chucked and picked up.
  MOVES P1                    The robot moves to retract point 1.
  UNSYNC (1)                  Synchronization ends.
  IGNORE XIN(1)
            :
  CONFIG-LEFTY                The left-shoulder system is selected for robot configuration.
  MOVE P2                     The robot moves to waiting point 2.
  SYNC (2)                    Waiting for LS2 ON.  With LS2 ON, synchronization starts.
            :                 Workpiece is placed and unchucked on conveyor 2.
  MOVES P2                    The robot moves to retract point 2.
  UNSYNC (2)                  Synchronization ends.
  IGNORE XIN(2)
            :
  GOTO L100
L200:
  PRINT "C/N ERROR=", N, CR   Error is displayed on the teach pendant.
  STOP                        Program execution stops.
END
```

```
DATA
  A1 = ~                    Conveyor 1 synchronization start point
  B1 = ~                    Conveyor 1 synchronization end point
  P1 = ~                    Conveyor 1 waiting point/retract point
     :                      Conveyor 1 ..... .
  A2 = ~                    Conveyor 2 synchronization start point
  B2 = ~                    Conveyor 2 synchronization end point
  P2 = ~                    Conveyor 2 waiting point/retract point
     :                      Conveyor 2 ..... .
END
```

- A workpiece is chucked and picked up from conveyor 1, and placed and unchucked on conveyor 2.
- Just after parameter definition, each encoder cable is checked for breakage.

7. Operation

This section describes the operating procedures for the conveyor synchronization function.

7.1 Stop Operation

The following stop operations are available for the TS2000 controller.

- STOP command and external stop signal input
- BREAK command
- Master mode switch
- STOP command of SCOL language
- MODE selection (STEP/MOTION)
- FIELD HOLD command

} A. Stop with servo ON

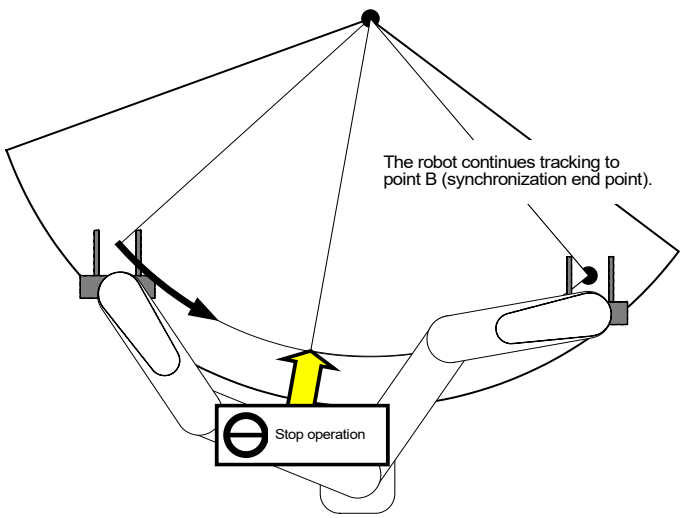
- Emergency stop
- Servo OFF (external servo OFF input signal)

} B. Stop with servo OFF

When the stop operation is executed during conveyor synchronization (i.e., while the robot is moving by tracking the conveyor speed), the results are as follows:

A: Stop with servo ON

The program stops, but the robot will not stop and continues tracking the conveyor to [B: Conveyor synchronization end point].



If the robot is kept intact under this condition, the robot stops at [B: Conveyor synchronization end point]. As a result, the robot hand may collide with a workpiece on the synchronized conveyor. To prevent this, the system should be constructed, fully considering the following matters.

[1] If the conveyor or other external equipment enters an emergency stop state, the robot should also enter an emergency stop state.

[2] If the robot enters an emergency stop state, the conveyor or other external equipment should also enter an emergency stop state.

[3] If the robot program stops due to some reason (error stop/stop operation by operator), the conveyor should be stopped also.

B: Stop with servo OFF

The robot stops instantaneously to stop the synchronization processing instantly.

When performing the stop operation during conveyor synchronization, strictly observe the following precautions.

CAUTION

- Even if the program stopped during conveyor synchronization, continuous operation is possible by using the restart function just after the program stop. In principle, however, after the stop operation, be sure to reset the operation, move the robot to the safe place by manual guidance, reset the program, then resume the operation. (For details, see Para. 7.2.)
- Unless an error is caused, DO NOT effect the servo OFF or emergency stop during conveyor synchronization. Otherwise, the robot hand will collide with a workpiece on the synchronized conveyor.
- To avoid the above accident, construct the system, taking notice of the following matters.

  [1] If the conveyor or other external equipment enters an emergency stop state, the robot should also enter an emergency stop state.

  [2] If the robot enters an emergency stop state, the conveyor or other external equipment should also enter an emergency stop state.

  [3] If the robot program stops due to some reason (error stop/stop operation by operator), the conveyor should be stopped also.

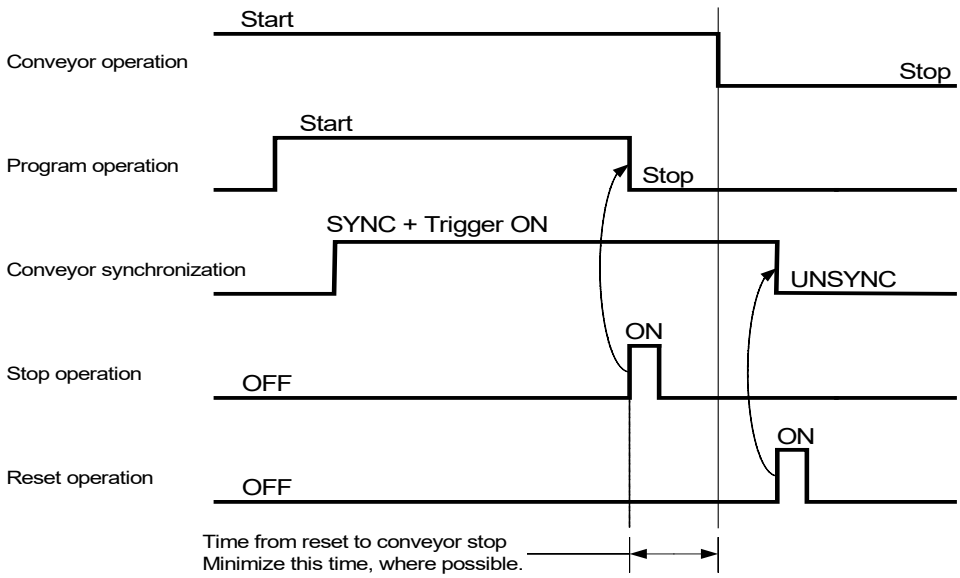
7.2 Reset Operation

When the reset operation is executed after the stop operation done during conveyor synchronization, the processing of conveyor synchronization stops. In other words, the UNSYNC command (i.e., synchronization end command) is executed by the following reset operation.

The reset operation comes in the following three (3) types.

[1] Program reset
[2] Step reset
[3] Cycle reset

After the stop operation as described in Para. 7.1 is executed during conveyor synchronization, be sure to perform the reset operation before restarting the program. When executing the reset operation, however, be sure to strictly observe the following procedures. Otherwise, the robot hand may collide with a workpiece on the synchronized conveyor.



Cautions:

- After the stop operation and before the reset operation, be sure to stop the conveyor. The time covering from the reset operation to the conveyor stop as indicated in the above figure should be as short as possible.

- If the stop operation in Para. 7.1–B has been effected, the time covering from the reset operation to the conveyor stop as indicated in the above figure should be almost zero (0).

7.3 Conveyor Stop

If the number of encoder pulses counted is zero (0) during the time designated by the construct parameter after the start of conveyor synchronization following the synchronization trigger signal ON, the error saying "001–048: Conveyor Stop" generates. This is the error of level 1, and the program is executed continuously without being stopped.

The conveyor (encoder) speed fluctuation should be within ±7.5 % to the average constant speed of the conveyor being used. Otherwise, precise synchronous accuracy cannot be assured, and the workpiece or hand, etc., may be damaged accordingly.

7.4 Conveyor Synchronization Utility (Utility → Conv)

This paragraph describes the utility function for performing various monitoring operations of the conveyor synchronization function.

```
CONV UTILITY      <CH1>      <CH2>
  Conv Form    :   LINEAR     ARC(+)
  Conv Veloc   : -****.***  -****.****
  Conv Status  :         0          0
  Enc Pulse    :     32767      32767
  Mes Pulse    :     63241          0
  Mes Trip     : -****.***  -****.****
  MES1   MES2
```

[1] Conveyor trigger signal monitor:

: Inversely displays <CH1> or <CH2> when the conveyor trigger signal has turned on.

<CH1>: Trigger signal OFF

<CH1>: Trigger signal ON

[2] Conv Form : Indicates the conveyor synchronization method.

LINEAR → Linear

ARC (+) → Arc (CW)

ARC (–) → Arc (CCW)

[3] Conv Veloc : Indicates the conveyor speed.

[4] Conv State : Indicates the conveyor state.

0 : During asynchronization

1 : Waiting for synchronization signal

2 : During synchronization delay

3 : During acceleration for synchronization

4 : During constant-speed for synchronization

5 : During deceleration for synchronization

6 : During processing of synchronization end

–1 : Conveyor breakage

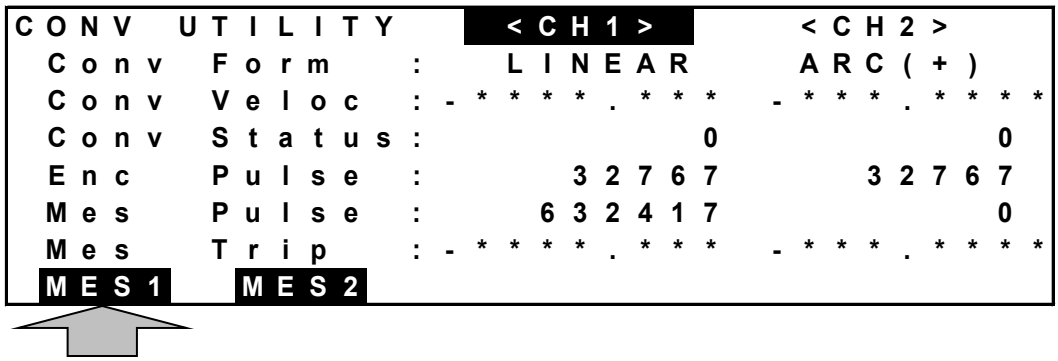
[5] Enc Pulse : Indicates the pulse counter value of the conveyor encoder.

[6] Mes Pulse : Indicates the measured value of total pulses of the conveyor encoder.
When you press the <MES1> (F1) or <MES2> (F2) function key, measurement of total encoder pulses starts.
For the measuring procedures, see Para. 1.2.1.

[7] Mes Trip : Indicates the calculated value of conveyor travel distance.
The value of "Mes Pulse" multiplied by the travel distance per pulse of the encoder, which is registered beforehand in user parameter U20, is indicated.

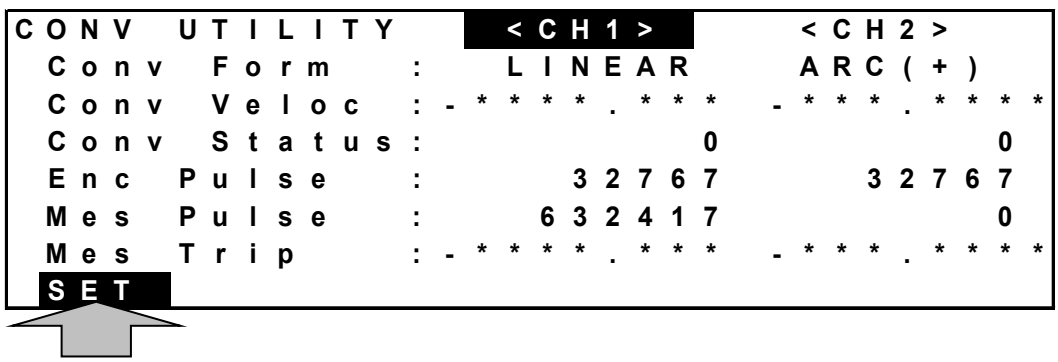
7.4.1. Measuring Total Pulses of Conveyor Encoder

(1) Press the <MES1> (F1) or <MES2> (F2) function key corresponding to the channel (CH1 or CH2) you wish to measure, using the cursor.

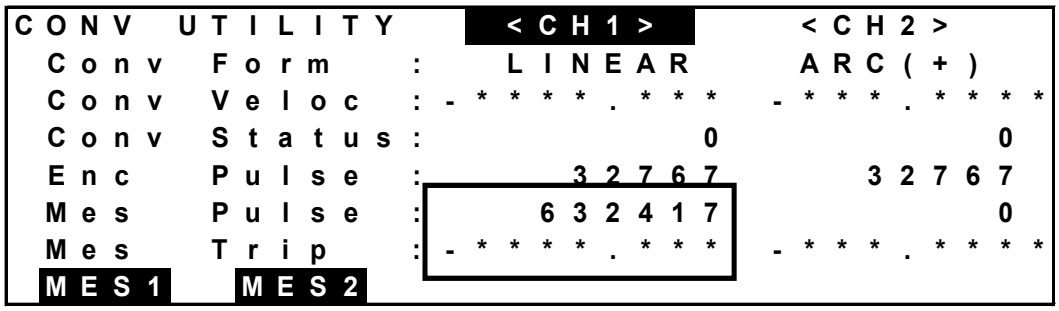


When you wish to count the number of pulses of CH1, press the <MES1> (F1) function key.

(2) When you wish to count the number of pulses of CH1, press the <MES1> (F1) function key. Likewise, when you wish to count the number of pulses of CH2, press the <MES2> (F2) function key. After the conveyor has been moved by "ΔL", press the <SET> key.



(3) When you press the <SET> key, the number of pulses measured "Mes Pulse" and the value which was calculated by converting the value of "Mes Pulse" into the travel distance in unit "Mes Trip" are displayed on the screen.
To measure again, press the <MES1> (F1) or <MES2> (F2) key and repeat the above steps (1) and (2).



Caution: The value of "Mes Pulse" cannot be displayed correctly unless the conveyor parameter has been set. To set this parameter, the travel distance per pulse should be figured out, using the value of "Mes Pulse" measured and the value of the conveyor move distance (actually measured value), which should be set then in [U20] Conveyor Parameter. For the parameter setting procedures, see Para. 1.1 [8].

8. Troubleshooting

8.1 Error Message Table

If an error has been detected by the self-diagnosis function, appropriate alarm message is displayed on the teach pendant, as shown below. Identify the cause and take appropriate measures according to the following table.

Error code	Error message	Possible causes and remedies
001–047	Conveyor Line Breakage	• The encoder cable of the specified conveyor number is broken or the encoder is defective. Check for the wiring. • Even after this error occurred, the robot continuously tracks the conveyor based on the conveyor (encoder) speed just before the error generation, without being stopped. • If the encoder cable breakage has been detected during conveyor synchronization, be sure to recess the robot to the safe place. Otherwise, the robot may collide with the workpiece or conveyor. • This error will not be reset unless the power is turned off. Locate and remedy the broken part, then turn the power on again.
001–048	Conveyor Stop	• The conveyor stop time has reached the judgment time as specified by the parameter. • The conveyor was stopped when the UNSYNC command has been executed.
002–033	Conveyor Parameter Error	• The parameter value exceeds the permissible range.

8.2 Troubleshooting

The error messages originated from the processing of conveyor synchronization are listed below. If the step number of the position where an error occurred indicates the CONV, SYNC, or UNSYNC command, identify the cause and take appropriate measures according to the following table.

Command	Error message and possible causes
CONV	002–042: Data Type error • The type of the variable used in the argument is illegal.
	002–025: Invalid command • The conveyor synchronization function is not operative.
	002–033: Conveyor Parameter Error • The value exceeds the permissible range.
SYNC	002–041: Invalid Variable • The value exceeds the permissible range.
	002–025: Invalid command • The conveyor synchronization parameter for <Number; N> is not defined. • The programming steps are illegal.
UNSYNC	002–041: Invalid Variable • The value exceeds the permissible range.
	002–025: Invalid command • The conveyor synchronization parameter for <Number; N> is not defined. • The conveyor synchronization has not started.
XIN	002–042: Data Type error 002–025: Invalid command • The number exceeds the permissible range.
Other	002–025: Invalid command error • An illegal motion control command is described between the SYNC command and UNSYNC command.